Corticocortical Projections to Representations of the Teeth, Tongue, and Face in Somatosensory Area 3b of Macaques

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ABSTRACT

We placed injections of anatomical tracers into representations of the tongue, teeth, and face in the primary somatosensory cortex (area 3b) of macaque monkeys. Our injections revealed strong projections to representations of the tongue and teeth from other parts of the oral cavity responsive region in 3b. The 3b face also provided input to the representations of the intraoral structures. The primary representation of the face showed a pattern of intrinsic connections similar to that of the mouth. The area 3b hand representation provided little to no input to either the mouth or the face representations. The mouth and face representations of area 3b received projections from the presumptive oral cavity and face regions of other somatosensory areas in the anterior parietal cortex and the lateral sulcus, including areas 3a, 1, 2, the second somatosensory area (S2), the parietal ventral area (PV), and cortex that may include the parietal rostral (PR) and ventral somatosensory (VS) areas. Additional inputs came from primary motor (M1) and ventral premotor (PMv) areas. This areal pattern of projections is similar to the well-studied pattern revealed by tracer injections in regions of 3b representing the hand. The tongue representation appeared to be unique in area 3b in that it also received inputs from areas in the anterior upper bank of the lateral sulcus and anterior insula that may include the primary gustatory area (area G) and other cortical taste-processing areas, as well as a region of lateral prefrontal cortex (LPFC) lining the principal sulcus. J. Comp. Neurol. 522:546–572, 2014.

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INDEXING TERMS: somatosensory cortex; gustatory cortex; taste; insular cortex; motor cortex

Primates have a large representation of the oral cavity, especially of the teeth and tongue, in the primary somatosensory cortex (area 3b). Unlike the rest of the body representation in 3b, this region contains a large bilateral representation of intraoral structures in both cerebral hemispheres (Manger et al., 1996; Jain et al., 2001a; Qi and Kaas, 2004). Despite its large size, functional importance in evaluating and processing food (Shepherd, 2012), expansion after spinal cord injury (Pons et al., 1991; Jain et al., 1997, 1998b, 2008; Fang et al., 2002), and role in coordinating movements of the tongue and jaw (Lin et al., 1993, 1994), little is known about the connections of the cortical representation of the oral cavity in macaques. This is due, in part, to the difficulties in reaching this most lateral part of S1 during microelectrode mapping experiments and studies of connections based on cortical injections.

Much like the hand, representations of different parts of the oral cavity and face have been found in multiple areas of somatosensory and motor cortex. Microelectrode mapping studies have shown, at least in part, orderly arrangements of the responses to stimulation of the oral cavity and face in somatosensory areas including 3b, 3a, 1, 2, the second somatosensory area (S2), and the parietal ventral area (PV) for Old World macaques (Nelson et al., 1980; Krubitzer et al., 1995; Manger et al., 1995, 1996; Toda and Taoka, 2001, 2002a,b, 2004; Qi and Kaas, 2004) and New World monkeys (Krubitzer and Kaas, 1990; Jain et al., 2001a; Qi et al., 2002; Iyengar et al., 2007). Representations of the face have been described for additional somatosensory areas in the lateral sulcus and the ventral

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somatosensory (VS) and parietal rostral (PR) areas for New World titi monkeys (Coq et al., 2004). The existence of similar representations has been suggested for macaques (Krubitzer et al., 1995), although mapping in these regions has been incomplete. In the frontal lobe, intracortical stimulation (McGuinness et al., 1980; Ghosh and Gattera, 1995; Graziano et al., 2002), recording (Hoffman and Luschei, 1980; Murray and Sessle, 1992; Avivi-Arber et al., 2011), and cooling (Murray et al., 1991) have all indicated that regions of the lateral primary motor cortex (M1) and ventral premotor area (PMv) are involved in controlling movements of the face and tongue. In macaques, little is known about how these multiple mouth and face representations interconnect.

Most of the anatomical information about the cortical representation of the oral cavity has come from studies in New World monkeys, in which this cortex is most accessible. However, projections from within area 3b and the presumptive face representations in areas 3a, 1, 2, and 5 to representations of the upper face and lip in area 3b were described as being similar to those going to other parts of area 3b during a larger study of connections of areas in the anterior parietal cortex in macaques (Burton and Fabri, 1995). The lateral part of area 3b that is responsive to tactile stimulation of the mouth and face has been shown to overlie densely myelinated ovals lateral and anterior to the 3b hand representation in prosimian galagos and anthropoid marmoset, owl, squirrel, and macaque monkeys (Jain et al., 1998a, 2001a; Qi and Kaas, 2004; Kaas et al., 2006; Iyengar et al., 2007). Iyengar and colleagues (2007) used physiological mapping to target representations of individual intraoral structures, parts of the tongue, and upper and lower teeth, in primary somatosensory cortex for injection of neuroanatomical tracers in three species of New World monkeys. In all three species, the oral cavity representation in area 3b received patterns of projections from other somatosensory and motor areas that were largely similar to projections to other parts of 3b (Iyengar et al., 2007). Interestingly, injections into the representation of the tongue also labeled regions of cortex in the anterior upper bank of the lateral sulcus and insula that included the primary gustatory cortex (area G; Sanides, 1968; Iyengar et al., 2007) and possibly other areas involved in processing taste (Ogawa et al., 1989; Plata-Salamán and Scott, 1992; Scott and Plata-Salamán, 1999; Pritchard and Norgren, 2004). Projections of taste areas to the 3b representation of the tongue have not been described for macaques.

For the present study, we injected anatomical tracers into physiologically defined tongue, teeth, and face representations in area 3b to reveal corticocortical inputs to this region. This allowed us to compare the pattern of projections to the 3b oral cavity and face representations to those identified as going to other body part representations of area 3b, particularly the large and well-studied representation of the hand. Furthermore, these experiments allowed us to evaluate the possibility that intrinsic connections between hand and face representations in area 3b underlie the reactivation of hand cortex by inputs from the face after loss of sensory input from the hand (Pons et al., 1991; Jain et al., 1997, 1998b, 2008; Fang et al., 2002). In addition, our injections of retrograde tracers allowed us to use the locations of back-filled cells to determine the locations and extents of mouth and face representations in other cortical areas. Finally, our injections in the primary somatosensory tongue representation allowed us to provide evidence for the integration of touch and taste by identifying areas likely involved in processing taste in the opercular and insular cortex (Scott et al., 1986; Ogawa et al., 1989; Smith-Swintosky et al., 1991; Scott and Plata-Salamán, 1999; Pritchard and Norgren, 2004) that project to area 3b.

Abbreviations

AS	Arcuate sulcus
CS	Central sulcus
G	Gustatory cortex
ILS	Inferior limiting sulcus
INS	Insula
IOS	Inferior occipital sulcus
ipCS	Inferior precentral sulcus
ipoCS	Inferior postcentral sulcus
IPS	Intraparietal sulcus
LBLS	Lower bank of the lateral sulcus
LPFC	Lateral prefrontal cortex
LS	Lateral sulcus
LuS	Lunate sulcus
M1	Primary motor area
OFC	Orbitofrontal cortex
PMd	Dorsal premotor area
PMv	Ventral premotor area
PPC	Posterior parietal cortex
PR	Parietal rostral area
PrCS	Precentral sulcus
PS	Principal sulcus
PV	Parietal ventral area
S2	Second somatosensory area
SLS	Superior limiting sulcus
STS	Superior temporal sulcus
UBLS	Upper bank of the lateral sulcus
VS	Ventral somatosensory area

MATERIALS AND METHODS

Four adult macaque monkeys (one *Macaca radiata*, case 1, and three *Macaca mulatta*) were used in this study. Many of the procedures were described in our companion study of thalamocortical connections in these monkeys (Cerkevich et al., 2013). The experimental procedures were approved by the Vanderbilt University Animal Care and Use Committee and adhered to National Institutes of Health guidelines. All surgical procedures were performed under aseptic conditions.

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Surgical procedures

The animals were anesthetized with an initial dose of ketamine hydrochloride (10–50 mg/kg, i.m.), and secured in a stereotaxic frame. A surgical level of anesthesia was maintained with an i.v. ketamine drip (4 mg/ml in sterile saline) and supplemental injections of xylazine (0.4 mg/kg, i.m.). Urethane (1.6 mg/kg, i.p.) was also given to maintain a stable anesthetic plane during terminal procedures. Heart rate, blood oxygen levels, and temperature were monitored throughout the mapping procedure. Each monkey was placed on a heating pad or under a heating lamp to maintain body temperature at 37°C. A local anesthetic, lidocaine hydrochloride, was applied to the ears and subcutaneous skin before the skin was incised to expose the skull. A head post was attached to a portion of the skull that did not overlie the region of interest in order to avoid interference from eye and mouth bars during mapping. A craniotomy was performed to expose the lateral half of the central sulcus. The dura was removed, and the exposed cortex was kept moist with regular application of sterile saline until the injections were completed, then covered with sterile silicone oil during the subsequent microelectrode mapping. Photographs of the surface of the brain were used to mark the placement of microelectrodes, the types of responses at different sites, and the locations of tracer injections and lesions using blood vessels and sulcal patterns as landmarks.

Multiunit mapping and injections

Neuroanatomical tracers were injected into physiologically defined representations of the tongue, teeth, gingiva, palate, buccal wall, lip, and chin that were identified by inserting low-impedance tungsten (1.0 mΩ at 1,000 Hz) or stainless-steel (1.0 mΩ at 1,000 Hz) microelectrodes into the cortex to record multiunit activity while the intraoral structures and face were stimulated with wood or glass probes. Our goal was to orient the mapping electrode so that it was perpendicular to layer 4, but this was not always possible because much of the mouth representation is on the caudal bank of the central sulcus. The electrode was advanced in 200–300 μm steps with a hydraulic Microdrive (David Kopf Instruments, Tujunga, CA) starting at a depth of 600–700 μm and continuing until responses were lost, sometimes as deep as 7,300 μm. Electrode penetrations were marked on high-resolution digital photographs of the brain. Receptive fields were outlined on standard drawings of a macaque face and the inside of the mouth. After the initial mapping revealed sites of representations of the tongue, teeth, and face, the electrode holder was removed, and a Hamilton syringe with a glass pipette tip attached was placed onto the stereotaxic arm. This ensured that the path of the syringe followed the same angle used for mapping on the way to the targeted location. Pressure injections of up to three (Fig. 1) of the following anatomical tracers were placed across the 3b representations of the mouth and face: 0.25–0.6 μl cholera toxin subunit B (CTB; 1% in distilled water, Sigma, St. Louis, MO; or Molecular Probes, Carlsbad, CA), 0.25–0.3 μl Fluororuby (FR; 10% in distilled water, Molecular Probes or Invitrogen, Carlsbad, CA), 0.01–0.04 μl wheat-germ agglutinin conjugated with horseradish peroxidase (0.2% WGA-HRP in distilled water; Sigma), and 0.4 μl biotinylated dextran amine (BDA; 10% in phosphate buffer; Molecular Probes or Invitrogen). In three animals (cases 1, 3, and 4), recording continued for up to 3 days immediately after the injections to provide a more extensive map of the oral cavity representation and allow time for tracer transport. In case 2, because of the use of a different set of neuroanatomical tracers, injections were placed during a short initial mapping procedure. After the injections, gel film was inserted to replace the opened dura, the craniotomy was closed with dental cement, the muscle was reattached, and the skin was sutured closed. The animal then recovered from anesthesia and was treated with prophylactic antibiotic and analgesics. After 2 weeks, the optimal amount of time for tracer transport, the cortex was again exposed for more extensive recording around the locations of the injections performed as in the other cases. Once electrophysiological mapping was complete in all cases, the animals were given a lethal injection of sodium pentobarbital (80 mg/kg) and subsequently perfused through the heart with phosphate-buffered saline (PBS; pH 7.4), followed by 2–3% paraformaldehyde in buffered saline and 2–3% paraformaldehyde with 10% sucrose in PBS.

The cortex was separated from the rest of the brain. The sulci were opened, and each hemisphere was flattened and blocked so that sections cut tangential to the surface could be mounted onto glass slides (Fig. 2). The main somatosensory blocks were cut along the arcuate sulcus (AS) from the lateral tip to the spur, across the central sulcus (CS) medial to the area 3b hand face border to the intraparietal sulcus (IPS), through the IPS and the caudal lateral sulcus (LS), along the fundus of the inferior limiting sulcus (ILS), and then back up to the lateral AS (Fig. 2, dotted line). These blocks included all of the anterior parietal and lateral sulcal somatosensory fields, the insula, much of the frontal operculum, and the entire lateral motor and premotor cortices. Blocks containing the rest of the ipsilateral area 3b and cingulate and contralateral

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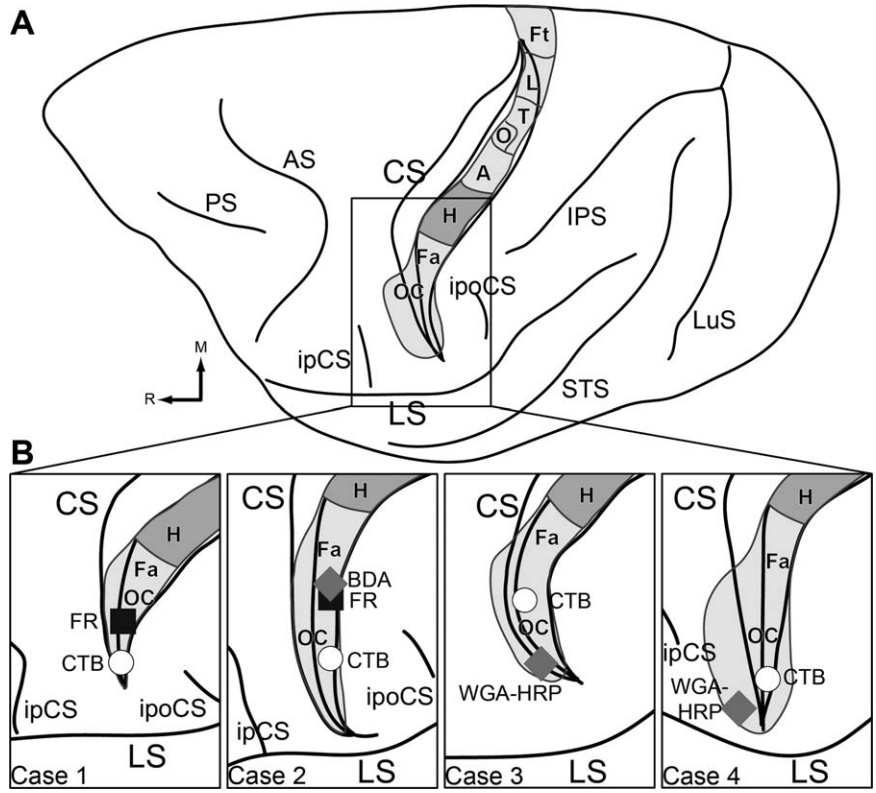


Figure 1. Locations of injection sites. **A:** Lateral view of the brain with the central sulcus opened. Area 3b is on the caudal bank of the central sulcus and shaded in light gray, with the hand representation in darker gray. The somatotopic organization in area 3b is indicated: OC, oral cavity; Fa, face; H, hand; A, arm; O, occiput; T, trunk; L, leg; Ft, foot (adapted from Nelson et al., 1980; Jain et al., 2008). **B:** Locations of injection sites in the open central sulcus of each case. Markers in the open central sulcus indicate the center of each tracer injection in the representations of the face and mouth of each case. For abbreviations see list. Rostral is left, medial is upward in both A and B.

somatosensory cortex were also flattened. The prefrontal and orbitofrontal cortex was either flattened or kept whole to be cut coronally. Flattened blocks were held between two glass slides and stored overnight in 30% sucrose at 4°C for cryoprotection, along with the unflattened blocks and thalamus from each case.

Cortical histology

The flattened blocks were cut tangential to the surface, whereas unflattened frontal blocks were cut coronally at a thickness of 40 or 50 μm on a freezing microtome. Series of sections were mounted unstained for fluorescence microscopy. Alternate series were appropriately processed to reveal tracers (Fig. 3). In all cases, one series was reacted for CTB immunohistochemistry (Bruce and Grofova, 1992; Angelucci et al., 1996) to visualize CTB-labeled cells. When appropriate, BDA was visualized by an avidin biotin-peroxidase reaction (ABC kit, Vectastain; Vector, Burlingame, CA; Veenman et al., 1992), and wheat germ agglutinin-horseradish peroxidase (WGA-HRP) series were processed with tetramethlybenzidine as chromogen and ammonium molybdate (Olucha et al., 1985) and stabilized in diamidinobenzidine (DAB). In all cases, series of sections were processed for myelinated fibers with both modified silver (Gallyas, 1979) and black gold (Schmued and Slikker, 1999) staining to reveal cortical myeloarchitecture.

Data analysis

Distributions of neurons filled with CTB, FR, WGA-HRP, or BDA were plotted with a fluorescent/brightfield Leitz microscope coupled to a computer running Neurolucida plotting software (MBF Bioscience, Williston, VT). The injection cores, the diffusion zones around them, and the regions of anterograde labeling revealed by bidirectional tracers were outlined. Landmarks including blood vessels, electrode tracts, and lesions were also marked for use during reconstruction.

Digital photomicrographs of sections stained to reveal architecture were taken with a Nikon DXM1200 camera mounted on a Nikon E800 microscope (Nikon,

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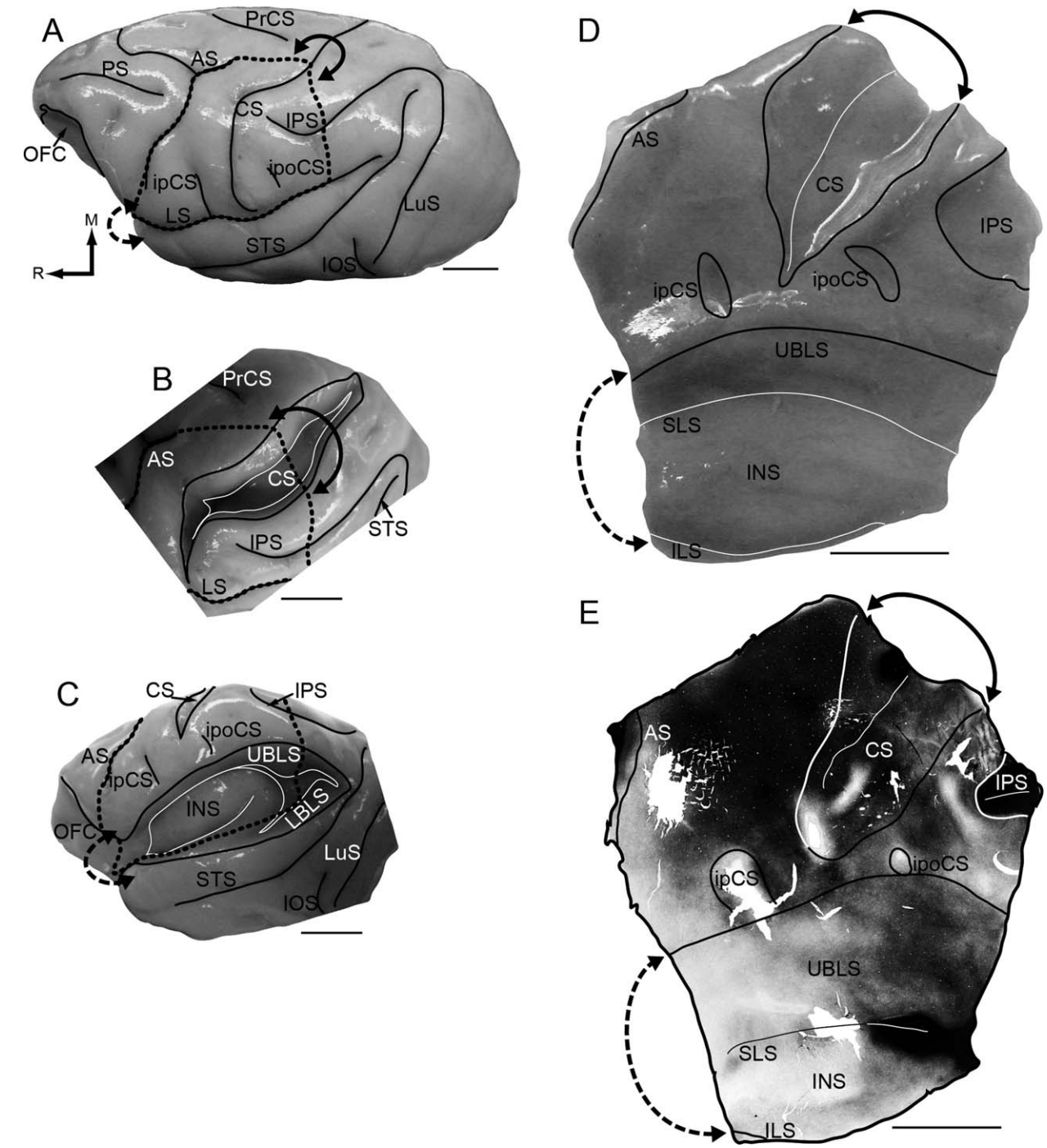


Figure 2. Cortical flattening procedure. **A:** Lateral view of the left hemisphere of case 2. **B,C:** Closer views of the opened CS (B) and LS (C). **D:** The flattened somatosensory block from the same hemisphere. **E:** Brightfield photomicrograph of a 40-μm-thick section through this block stained to reveal myelinated fibers, with sulci outlined. Myelinated fibers are stained in black, with the regions with the densest myelination staining most darkly. Heavy arrows indicate the opening of the central sulcus (CS, solid) and lateral sulcus (LS, dashed). Dashed lines indicate the region of the left hemisphere that was blocked to fit the size of the slides on which sections were later mounted. The block was cut to contain all of the area 3b oral cavity representation, the 3b hand–face border, digit representations and the lateral half of areas 1, 2, 3a, M1, and PMv, aand the cortex of the upper bank of the lateral sulcus and insula. For abbreviations see list. Rostral is left, medial is upward. Scale bars = 1 cm in A-C, 5 mm in D-E.

Melville, NY). These digital images were then adjusted for contrast, saturation, lightness, and curves with Adobe Photoshop CS2 (Adobe Systems, San Jose, CA) but were not otherwise altered. Photomicrographs were then stitched together in Adobe Photoshop to create an image of a single whole section as seen in Figure 2E.

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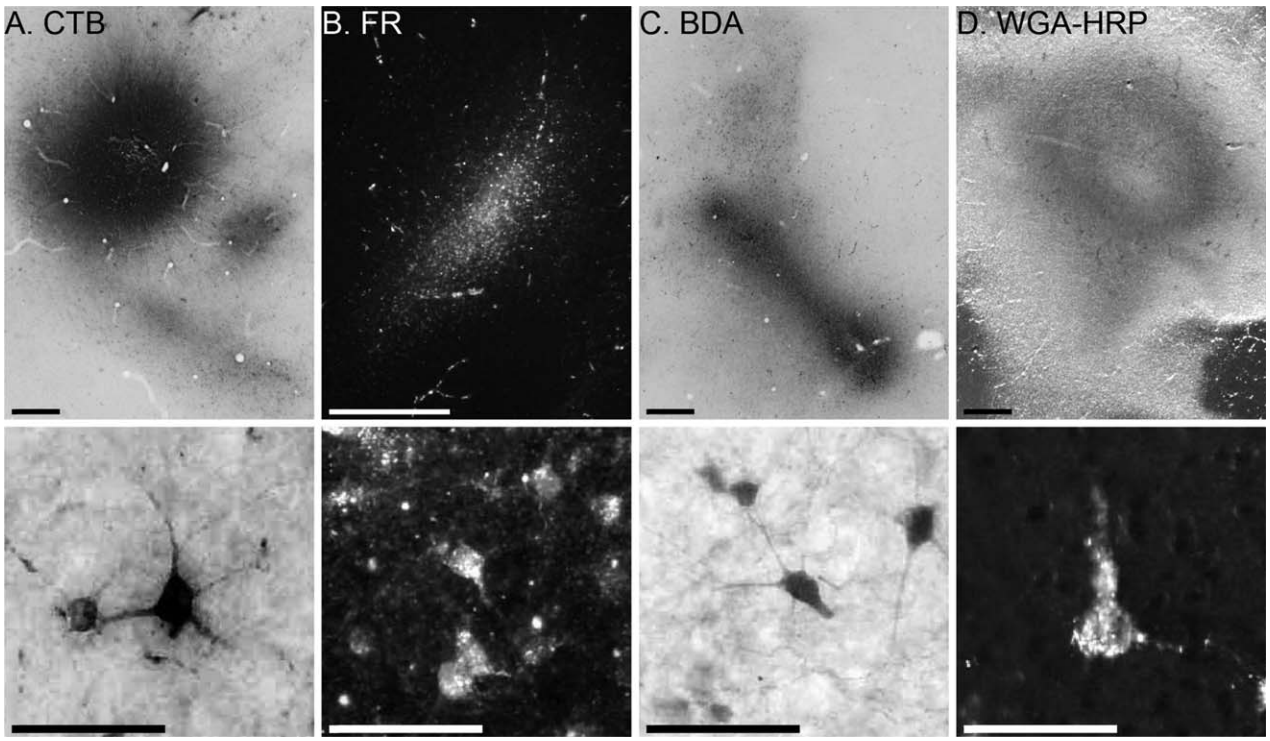


Figure 3. Tracer injection sites and tracer-filled neurons in cortex. Photomicrographs of tracer injection sites and tracer filled cells. Images were captured using a brightfield light microscope to show CTB (column **A**) and BDA (column **C**), a fluorescent microscope to show FR (column **B**), and under darkfield illumination to show WGA-HRP (column **D**). Scale bars = 0.5 mm in top row; 50 μ in bottom row.

Reconstruction

Cortical sections stained for myelinated fibers were used to determine the borders of cortical areas and to confirm the locations of injection cores and labeled neurons. Although dense myelin staining characterizes all of area 3b (Qi and Kaas, 2004), the face and mouth region of primary somatosensory cortex is separated from the rest of 3b by a myelin-poor septum known as the hand–face border (Jain et al., 1998a; Fang et al., 2002; Qi and Kaas, 2004). Myelin-dense ovals lateral to the hand–face border have been shown to indicate anatomical modules containing physiological representations of the different oral cavity and face structures in prosimian galagos and New World monkeys, and similar modules have been suggested for macaque monkeys (Jain et al., 2001b; Qi and Kaas, 2004; Kaas et al., 2006; Iyengar et al., 2007). Because the lateral extreme of the lateral sulcus is deep, curved, and often full of small folds that differ from animal to animal, complications of flattening and individual differences made it difficult to fully reveal the pattern of myeloarchitecture in this region of area 3b in macaques. However, we were able to find myelin-dense ovals representing the parts of the face and intraoral structures in each case (Fig. 4). Because the myelin ovals in macaques often appeared to be incomplete or disjointed because of imperfect flattening of cortex along the bank and base of the central sulcus, the myeloarchitecture within area 3b is best determined by reconstructing borders across serial sections through the depth of cortex.

To reconstruct the anatomical boundaries, the boundaries of myelin-dense regions of single sections were drawn using a projection microscope. A single section at the approximate depth of layer 4 (for example, Fig. 4B) was used as the reference section. The outline, blood vessels, and artifacts of the reference section were marked. The myeloarchitecture of area 3b, other areal boundaries, and indications of cortical sulci were then related to the reference section by using common landmarks, such as blood vessels, to align myelin-stained sections superficial and deep to the reference section (Fig. 4A and C, respectively). Physiological mapping data were added to the anatomical reconstructions by aligning penetrations along the trajectory of the electrode tracks that could be seen in the stained fiber sections and matching the locations of injection sites and lesions (see Fig. 5). Plotted cortex sections indicating the injection sites and distribution of back-filled cells were then aligned to the reconstructed borders and physiological maps using blood vessels, electrolytic

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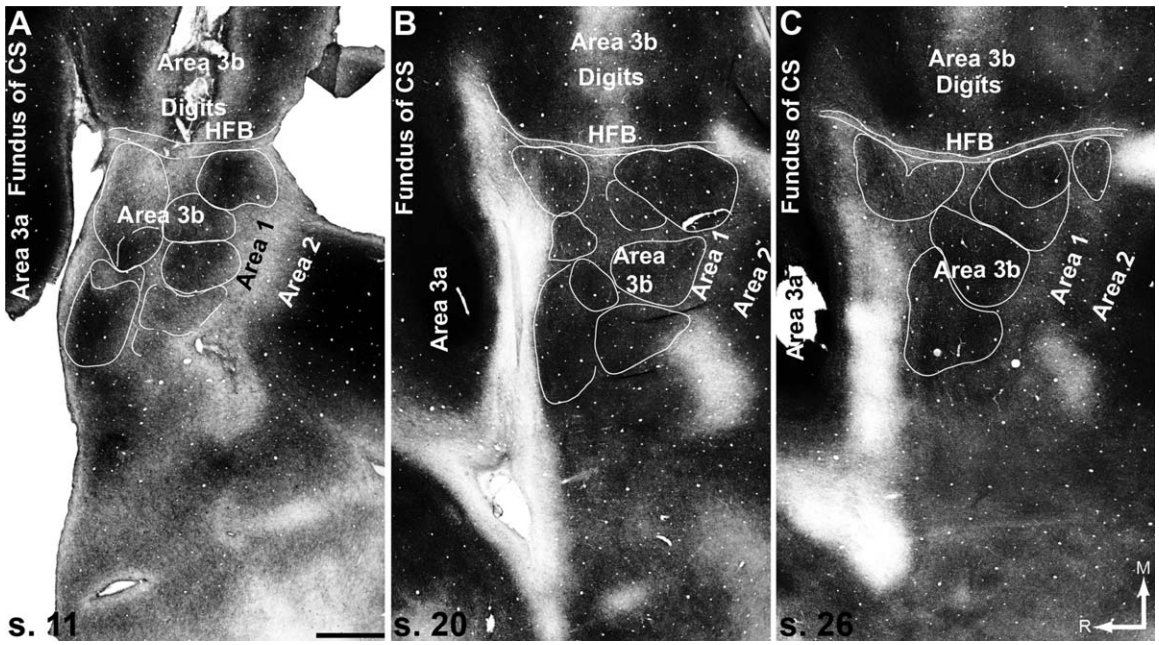


Figure 4. Myeloarchitecture of the 3b face and mouth region. **A–C:** Sections from case 4 stained to reveal myelinated fibers from most superficial (A) to deepest (C). Staining the cortex for myelin revealed dense patches that indicate the anatomical representation of the parts of the face and mouth through the depth of cortex in 3b oral cavity representation. These sections were used to draw a reconstruction of the areal boundaries throughout the depth of cortex by aligning corresponding landmarks, including blood vessels and electrolytic lesions from each section. White lines indicate the hand–face border (HFB) and myelinated ovals representing different parts of the face and mouth. These reconstructed borders were then aligned with physiological maps and plotted sections to indicate where labeled neurons were found in the brain. Sections were cut at 40 μm, and there are 320 μm between sections 11 and 20, and 200 μm between sections 20 and 26. For abbreviations see list. Rostral is left, medial is upward. Scale bar = 1 mm.

lesions, and the injection cores as landmarks. This allowed us to characterize the complete receptive field represented by the region encompassed by each injection. All reconstructions were created with Adobe Illustrator CS2 and CS5 (Adobe Systems).

RESULTS

Ipsilateral corticocortical connections

Case 1: tongue representation

The CTB injection was placed near the lateral tip of the central sulcus, into the region representing the tongue (Fig. 5A). Although it was centered on the representation of the tip of the tongue, the tracer diffused around the injection core to include all of the anterior tongue and much of the contralateral tongue (Fig. 6). This injection might have spread slightly into area 3a, although that could not be confirmed physiologically because we did not record from layer 4 in this location (Fig. 5A).

This injection revealed strong intrinsic connections within the area 3b tongue representation as well as with the representations of the other intraoral structures within area 3b, including the lower and upper teeth and tissue lining the inside of the cheek and lips (Fig. 6, Table 1). Representations of the lower lip and chin also contained cells labeled by the CTB injection into the region of 3b that responded to stimulation of the tongue. Filled neurons were also found in the 3b upper lip region. In contrast, the 3b hand representation was almost completely devoid of labeled cells (Table 2), indicating that there was little input from the hand region to the part of 3b serving the tongue.

Outside of primary somatosensory cortex, there were projections to the 3b tongue representation from other somatosensory areas in the anterior parietal cortex and upper bank of the lateral sulcus, cortex of the insula, and frontal cortex (Fig. 5). Though unmapped, regions of lateral areas 3a, 1, and 2 that likely contain representations of the inside of the mouth provided input to the 3b tongue. The field of labeled cells extended into lateral area 5 and the posterior parietal cortex (PPC). Back-filled neurons were found within the locations of the second somatosensory (S2) and parietal ventral (PV) areas, and they tended to be concentrated along the medial border of both areas, a region known to be responsive to stimulation of the face and mouth (Krubitzer et al., 1995). Separate foci of labeled cells could

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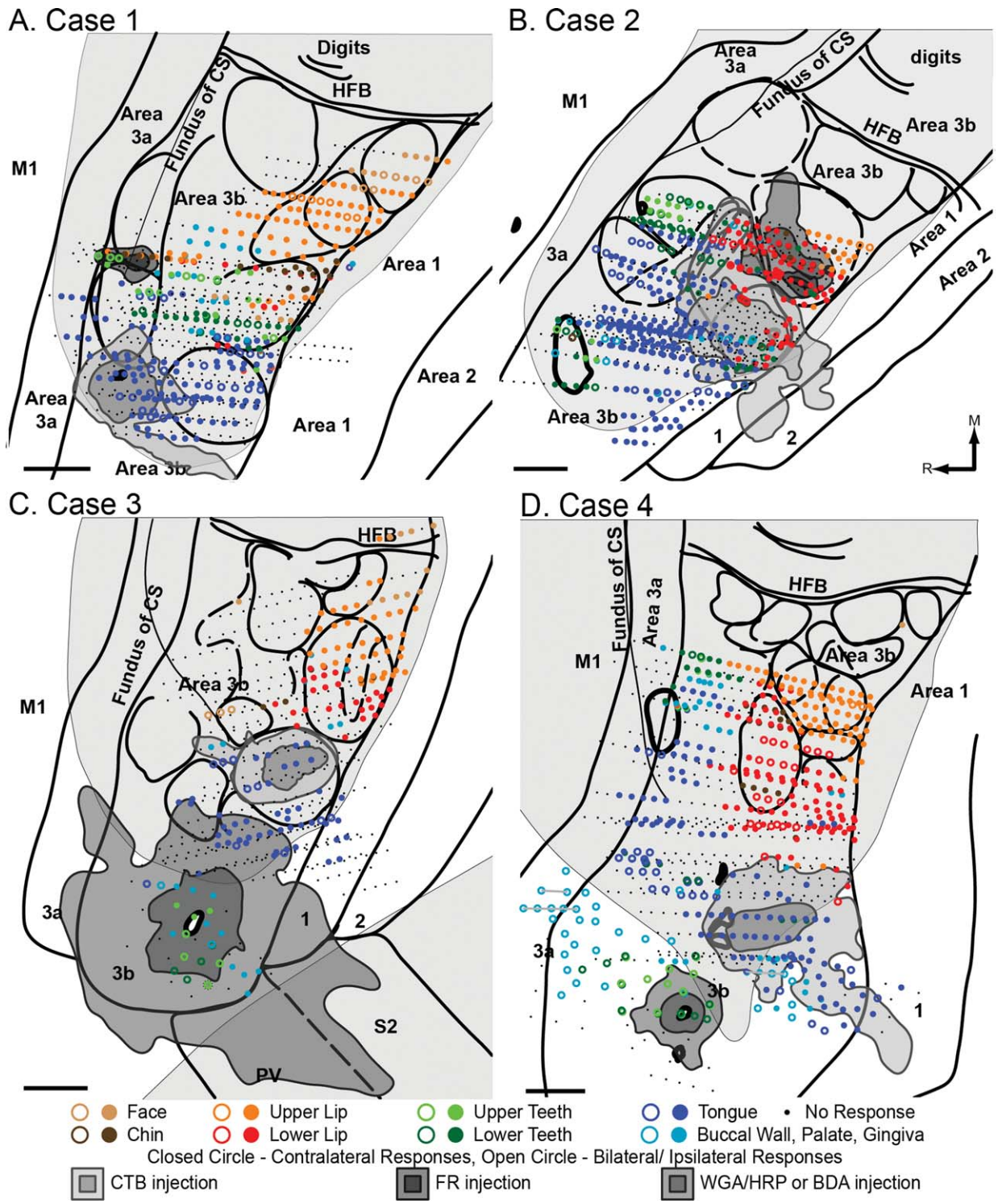


Figure 5. Physiological characterization of injection sites. **A–D:** Physiological maps of responses to somatosensory stimulation of the face and intraoral structures during microelectrode mapping of area 3b aligned to reconstructions of myeloarchitecture and injection sites for each case, respectively. Thin black lines indicate the myeloarchitecture within area 3b and areal boundaries. Thick black lines are tears. The CS and LS are shaded in light gray. Circles are color coded to indicate the area of skin stimulated to elicit responses at that cortical location during mapping. Injection cores and surrounding diffusion zones are outlined and shaded in darker grays. The anatomical and electrophysiological maps were aligned on the basis of landmarks including tracer injection sites, lesions, and electrode tracts. For abbreviations see list. Rostral is left, and medial is upward. Scale bars = 2 mm.

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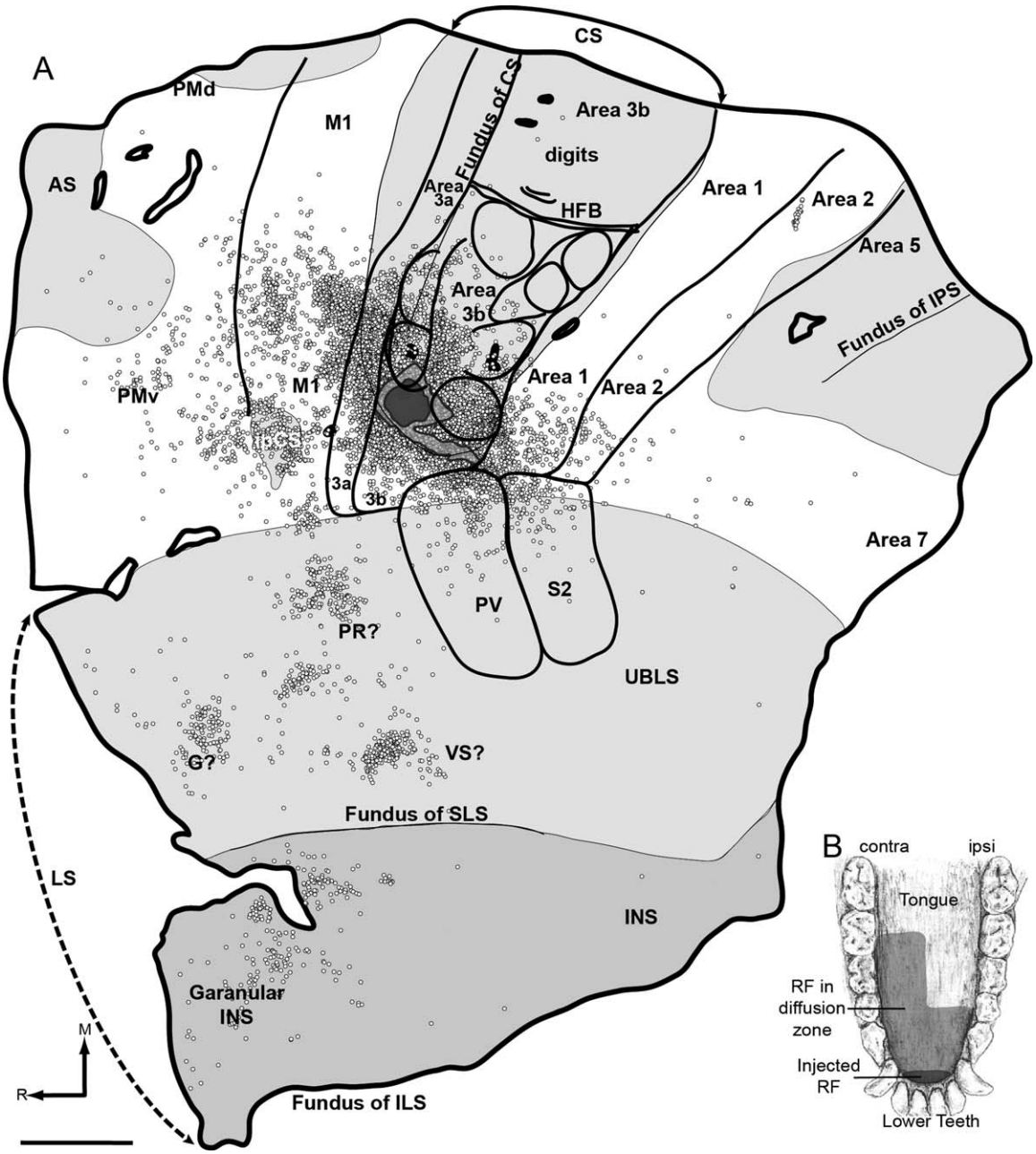


Figure 6. Case 1 CTB injection into 3b tongue. **A:** Distribution of CTB-labeled cells through the somatosensory cortex of case 1. Each white circle indicates the location of an individual CTB-filled cell. The core, full extent of diffusion zone around the injection, and patches of anterogradely filled axon terminals are outlined and shaded in dark gray. Thin black lines are anatomical borders. Thick black lines denote tears and the edges of the block. Large shaded regions indicate cortex in a sulcus; the fundus of each sulcus is marked by fine black lines. Areal boundaries were determined from adjacent sections stained for myelin, physiological mapping, and measurements based on previously published studies. This CTB injection into the 3b tongue representation revealed strong local connections with other parts of the 3b mouth and face regions, as well as inputs from other somatosensory areas in the parietal cortex and upper bank of the lateral sulcus, M1, PMv, and upper bank of the lateral sulcus and insular cortex that may be involved in taste processing. For abbreviations see list. Rostral is left, medial is upward. **B:** Injected receptive field. The darkly shaded region is the receptive field (RF) represented at the center of the injection. The lightly shaded region is an estimation of all parts of the mouth and face represented in the full extent of the tracer diffusion zone around the core of the injection. Scale bar = 5 mm.

also be found in the cortex rostral to PV. Although the locations of these foci were not identified architectonically, they presumably include face oral cavity representations in the parietal rostral (PR) and ventral somatosensory (VS) areas (Krubitzer et al., 1995; Qi et al., 2002; Coq et al., 2004). Lateral motor (M1) and

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TABLE 1.

Distributions (%) of Retrogradely Labeled Cells Throughout the Ipsilateral Cortex After Injections Into 3b Representations of the Oral Cavity and Face[1]

	Case 1		Case 2			Case 3		Case 4	
Area	Tongue (CTB; n = 18,447)	Teeth (FR; n = 2,758)	Tongue/lip (CTB; n = 34,128)	Lip (FR; n = 2,125)	Lip/chin (BDA; n = 7,264)	Tongue (CTB; n = 35,322)	Teeth (WGA-HRP; n = 8,967)	Tongue (CTB; n = 65,416)	Teeth (WGA-HRP; n = 5,255)
3b	62.54	84.52	54.32	96.05	96.83	61.69	35.51	63.18	86.41
3a	14.92	8.27	3.48	0.09	0.11	1.59	8.64	8.49	6.98
1	2.79	1.92	12.12	1.08	1.46	12.65	7.62	15.04	0.36
2	0.42	0.40	8.85	0.56	0.72	9.18	3.95	1.81	0.11
PPC	0.14	0.18	7.40	1.65	0.67	0.37	0.71	0.03	0.61
S2/PV	3.72	0.69	2.65	0.05	0.00	5.00	7.05	2.22	0.88
UBLS	3.95	3.30	2.49	0.00	0.06	2.47	20.56	3.18	1.08
INS	1.27	0.36	0.56	0.00	0.15	0.03	0.67	1.13	1.01
M1	8.61	0.33	6.38	0.52	0.00	4.73	9.90	4.85	1.39
PM	1.6	0.04	1.74	0.00	0.00	2.28	5.27	0.06	3.06
LPFC	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

[1]Rows sequentially list case numbers, injected representations, injected tracers, and total number of cells (n) labeled by each injection. Cortical areas containing labeled cells are listed in the first column, and successive columns contain the percentage of cells labeled in each area. PPC includes all parietal areas caudal to area 2. Because they were often difficult to distinguish, cells in area S2 and PV were counted together. UBLS includes all of the cells labeled in the upper bank of the lateral sulcus that were not in any part of anterior parietal areas 3b, 3a, 1, and 2 that extended onto the lateral sulcus or in S2/PV.

ventral premotor (PMv) each contained CTB-labeled cells. No motor mapping was performed for this study, but previous studies in motor and premotor cortex lateral to the arcuate sulcus have shown that microstimulation in both lateral M1 and PMv can evoke movements of the face and mouth (McGuinness et al., 1980; Qi et al., 2000; Graziano et al., 2002) and that this region is important to the proper performance of tasks involving movement of the tongue and jaw (Hoffman and Luschei, 1980; Murray et al., 1991; Murray and Sessle, 1992). Thus, we believe that our results show that regions of motor and premotor cortex involved in moving the tongue project to the 3b tongue representation. The injected 3b tongue representation also received projections from more anterior parts of the upper bank of the lateral sulcus and insula. Patches of filled neurons were found near the fundus of the superior limiting sulcus in the presumptive location of the primary gustatory area (G) and other taste-related areas of the anterior insula (Sanides, 1968; Ogawa et al., 1989; Smith-Swintosky et al., 1991; Plata-Salamán and Scott, 1992; Ogawa, 1994; Pritchard and Norgren, 2004). A few cells (Table 1) labeled by this injection were also found in the lateral prefrontal cortex, near the fundus of the principal sulcus (not shown).

Case 1: medial representation of the teeth

An injection of FR in case 1 was placed near the fundus of the central sulcus into a region that responded to touches on all of the upper front teeth and contralateral premolars; the skin of the medial upper lip, proximal chin, and anterior neck; and the hairs on the neck and jaw (Fig. 5A). The contralateral upper canine and all four upper incisors were represented at the center of this injection (Fig. 7). Like the injection into the representation of the tongue in this case, this injection might have spread very slightly into area 3a.

TABLE 2.

Distributions (%) of Cells Labeled Within Area 3b[1]

	Case 1		Case 2			Case 3		Case 4	
Area	Tongue (CTB; n = 11,536)	Teeth (FR; n = 2,331)	Tongue/lip (CTB; n = 18,539)	Lip (FR; n = 2,041)	Lip/chin (BDA; n = 7,034)	Tongue (CTB; n = 21,789)	Teeth (WGA-HRP; n = 3,184)	Tongue (CTB; n = 41,327)	Teeth (WGA-HRP; n = 4,541)
3b Mouth and face	99.97	99.96	99.36	100	99.97	99.96	99.84	100	99.87
3b Hand	0.03	0.04	0.64	0.00	0.03	0.04	0.16	0.00	0.13

[1]Conventions as in Table 1. 3b Mouth and face includes all cells in area 3b lateral to the dorsal edge of the hand face border.

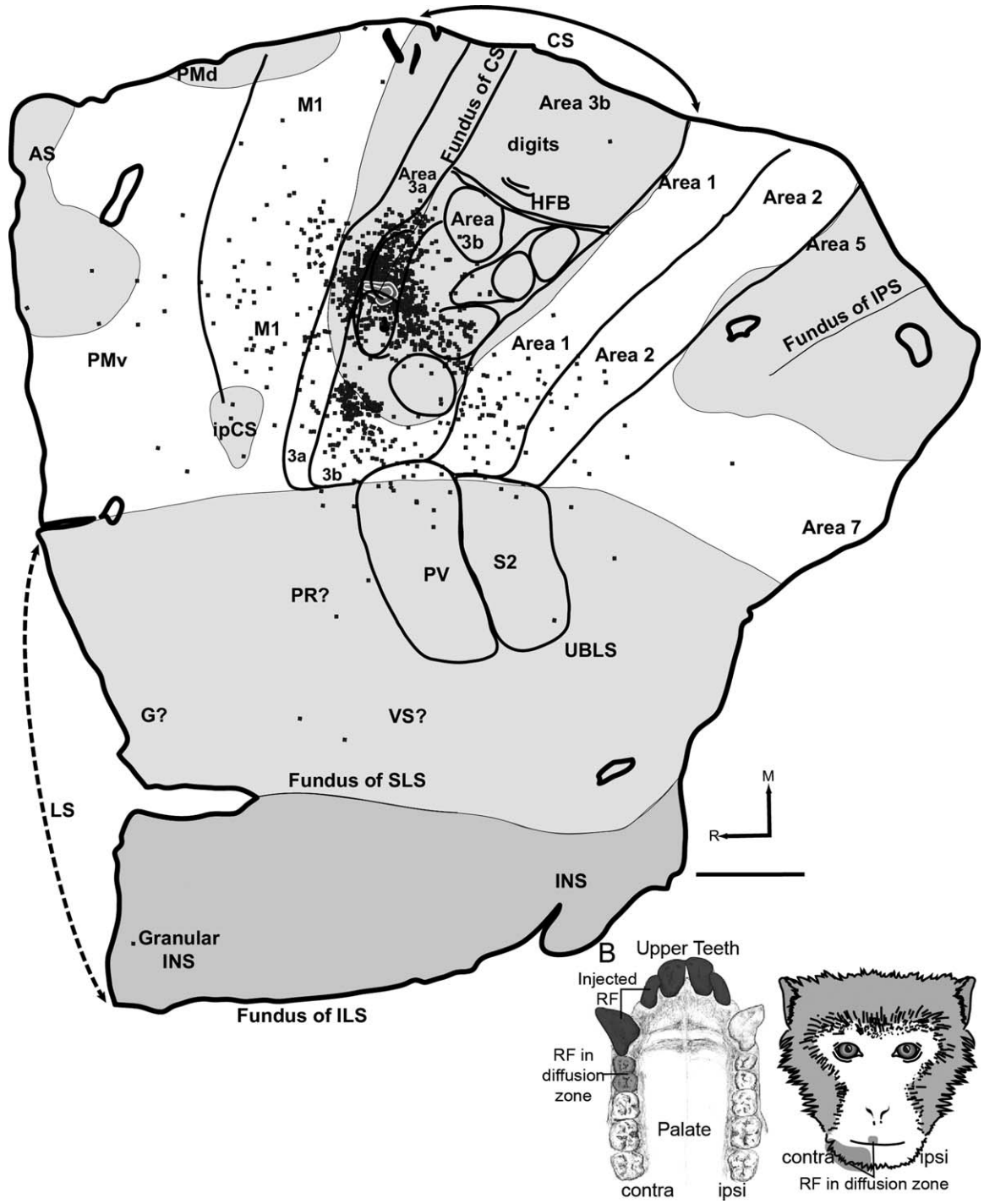


Figure 7. Case 1 FR injection into 3b teeth. **A:** Distribution of FR-labeled cells through the somatosensory cortex of case 1. Each black square indicates one FR-filled cell. The injection core and diffusion zone are outlined in white. An injection of FR into a region responding to the upper front teeth, lip, and chin resulted in filled cells scattered throughout somatosensory and motor cortex in a pattern similar to the case 1 tongue representation injection. There was, however, little projection from possible taste cortex in the insula and upper bank of the lateral sulcus to the representation of the teeth in this case. For abbreviations see list. Conventions as in Figure 6. **B:** Injected receptive field. Scale bar = 5 mm.

Again, the majority of inputs to the injected representation of the teeth came from other parts of the 3b representation of the mouth (Tables 1, 2), specifically the teeth, gums, and buccal wall (Fig. 7). A focus of labeled cells was also found in the rostrolateral part of area 3b that continued out of the central sulcus and onto the

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surface of the brain. This region was not mapped in this case for technical reasons, but, given the mapping results in our other cases (Fig. 5), we believe that these cells likely lie in a second, more lateral representation of the teeth and gums that includes more ipsi- and bilaterally responsive sites than the injected medial representation of the teeth. The adjacent tongue representation also provided inputs to the representation of the teeth and lip, although perhaps not from the center of the tongue region that was injected with CTB. The lip and chin representations in 3b projected to the adjacent medial representation of the teeth as well. There was almost no input from the 3b hand representation (Table 2).

Cells filled by the FR injection were also located outside of area 3b, in a pattern that was largely similar to that revealed by the CTB injection into the tongue region in case 1 (Fig. 7). Labeled cells were in presumably matched representations of the teeth in areas 3a, 1, 2, 5, S2, and PV. A few back-filled neurons were found anterior to PV in the lateral sulcus. M1 and PMv also contained labeled cells. Unlike the tongue representation injection, neurons were not labeled in regions of cortex that may process taste by our injection into the more medial representation of the teeth.

Case 2: tongue and lip representation

In case 2, an injection of CTB was placed on the caudal bank of the central sulcus into a region representing the contralateral anterior tongue, including the tip (Fig. 5B). A second mapping session revealed that the center of this injection included responses to touch on the whole of the contralateral dorsal surface of the tongue, the lower contralateral anterior teeth and ipsilateral incisors, part of the anterior palate, and skin and hairs of the contralateral lower lip and chin (Fig. 8). This injection spread to include 3b representations of the gums around the contralateral anterior lower teeth and possibly the ipsilateral first and second premolars (very weak responses to stimulation of these teeth were indicated). The diffusion zone around this CTB injection might also have included parts of adjacent areas 1 and 2 (Fig. 4B), but strong uptake from the diffusion zone was unlikely.

The injection centered on the tongue and lip representations in case 2 revealed a pattern of projections similar to that seen in case 1 (Fig. 8, Table 1). Labeled cells were found in other parts of 3b responding to stimulation of the tongue, lips, and chin. Within 3b, inputs to the tongue/lip representation also came from the more medial representation of the teeth, although a few labeled cells were found in the electrophysiologically defined lateral representation of the teeth as well. There were labeled cells in an unmapped portion of area 3b between the anatomical hand–face border and mapped lip and teeth representations that likely represents other parts of the face (Qi and Kaas, 2004; Kaas et al., 2006). Once again, the vast majority of filled neurons after the CTB injection into the 3b tongue/lip representation were in the region area 3b devoted to the face and mouth (Table 2). Outside of area 3b, there were back-filled cells in other somatosensory areas, including areas 3a, 1, 2, 5, S2, and PV. Labeled neurons were also found in the PPC and in the presumed locations of mouth and face representations in PR and VS. M1 and PMv also provided input to the injected tongue/lip region. The anterior upper bank of the lateral sulcus and insula contained CTB-labeled neurons, suggesting that taste cortex in these regions provides inputs to the 3b tongue representation.

Case 2: lip and chin representations

The case 2 BDA injection was our most medial injection (Fig. 5). It was centered on a region responding to both the skin and the hairs of the contralateral middle and medial lower lip and anterior chin (Fig. 9). This injection also involved a region responsive to all of the contralateral lower lip, including the corner of the mouth and the surrounding cheek and the rest of the chin. Some cortex representing the ipsilateral medial lower lip and upper anterior chin was included in this injection. The BDA and CTB injections overlapped slightly, with the BDA diffusion zone crossing into a region that included very weak responses to stimulation on the contralateral tongue.

The BDA injection in case 2 revealed a pattern of label similar to that seen after the case 2 CTB injection in the tongue/lip representation but with a more restricted distribution (Fig. 9, Table 1). Within area 3b, inputs to the chin/lip representation largely came from the rest of 3b representing the lips and chin. There were large numbers of back-filled cells in unmapped 3b cortex between the mapped representations of the chin and lips and the hand–face border that likely represent the rest of the face. Inputs from representations of the teeth came only from the nearby medial representation and not the more lateral part of 3b that was also responsive to touch on the teeth. A small, more medial part of the tongue representation also projected to the injected chin region. Because the BDA injection in chin and lower lip cortex was closest to the 3b hand–face border, and a chin representation has been show to appear in hand cortex after a spinal cord lesion damaged hand afferents (Manger et al., 1997), this injection was the best candidate to label latent connections between face and hand representations of area 3b.

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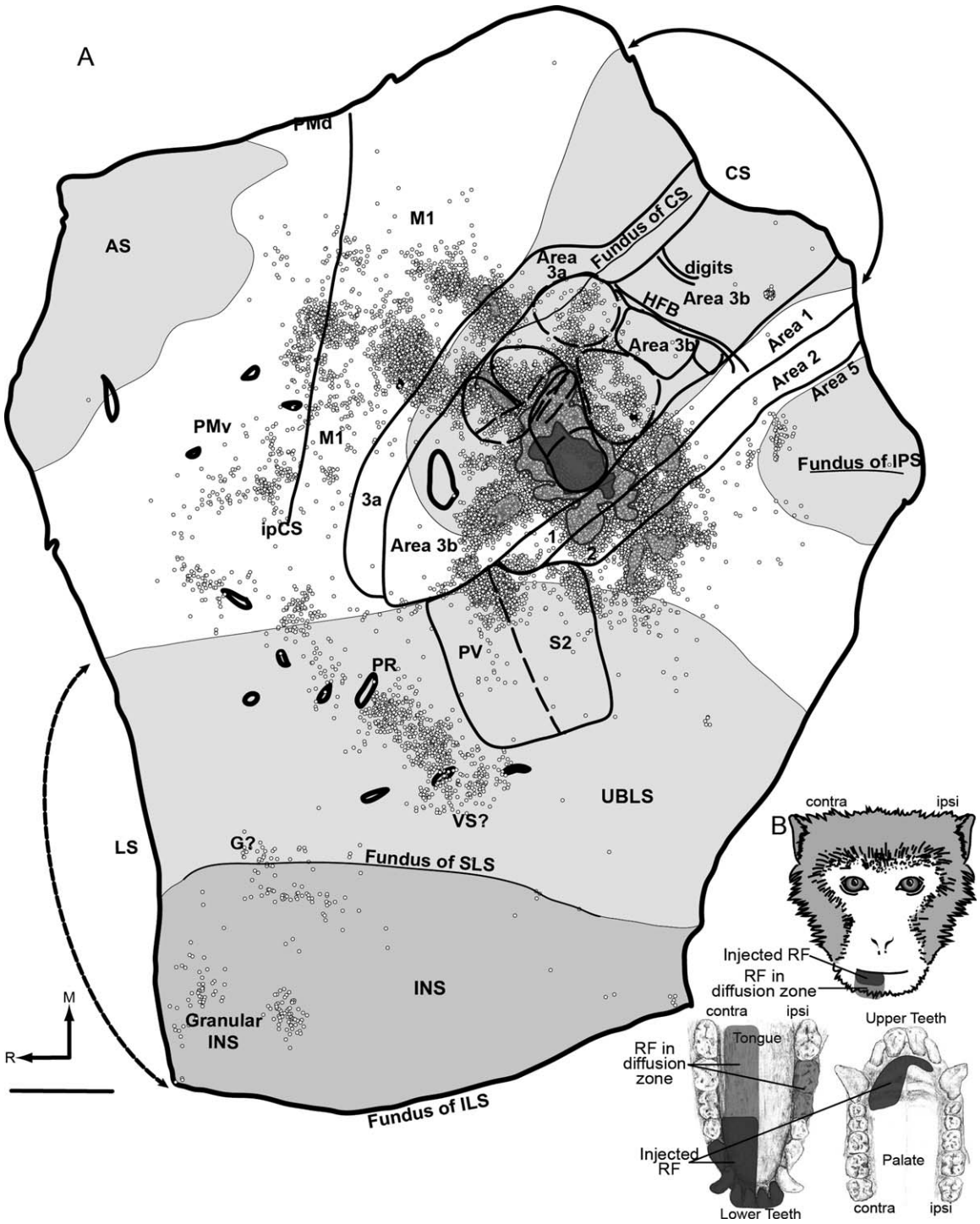


Figure 8. Case 2 CTB injection into 3b tongue/lip. **A:** Distribution of CTB-labeled cells through the somatosensory cortex of case 2. This injection of CTB includes parts of both the tongue and lip representations. Neurons labeled by this injection were found throughout the 3b face and mouth representation, in other anterior parietal and lateral sulcus somatosensory areas, in motor and premotor cortex, and in likely taste-responsive regions of the anterior upper bank of the lateral sulcus and insula. For abbreviations see list. Conventions as in Figure 6. **B:** Injected receptive field. Scale bar = 5 mm.

There was, however, no indication of cells projecting from the 3b hand cortex to the 3b face region (Table 2). A few cells within the septum between these two representations were filled with BDA. Outside of area 3b, the pattern of label was similar to that of the case 2 tongue/lip injection with inputs from all of the anterior parietal somatosensory areas indicated. Few labeled cells were found in M1 or the cortex of the

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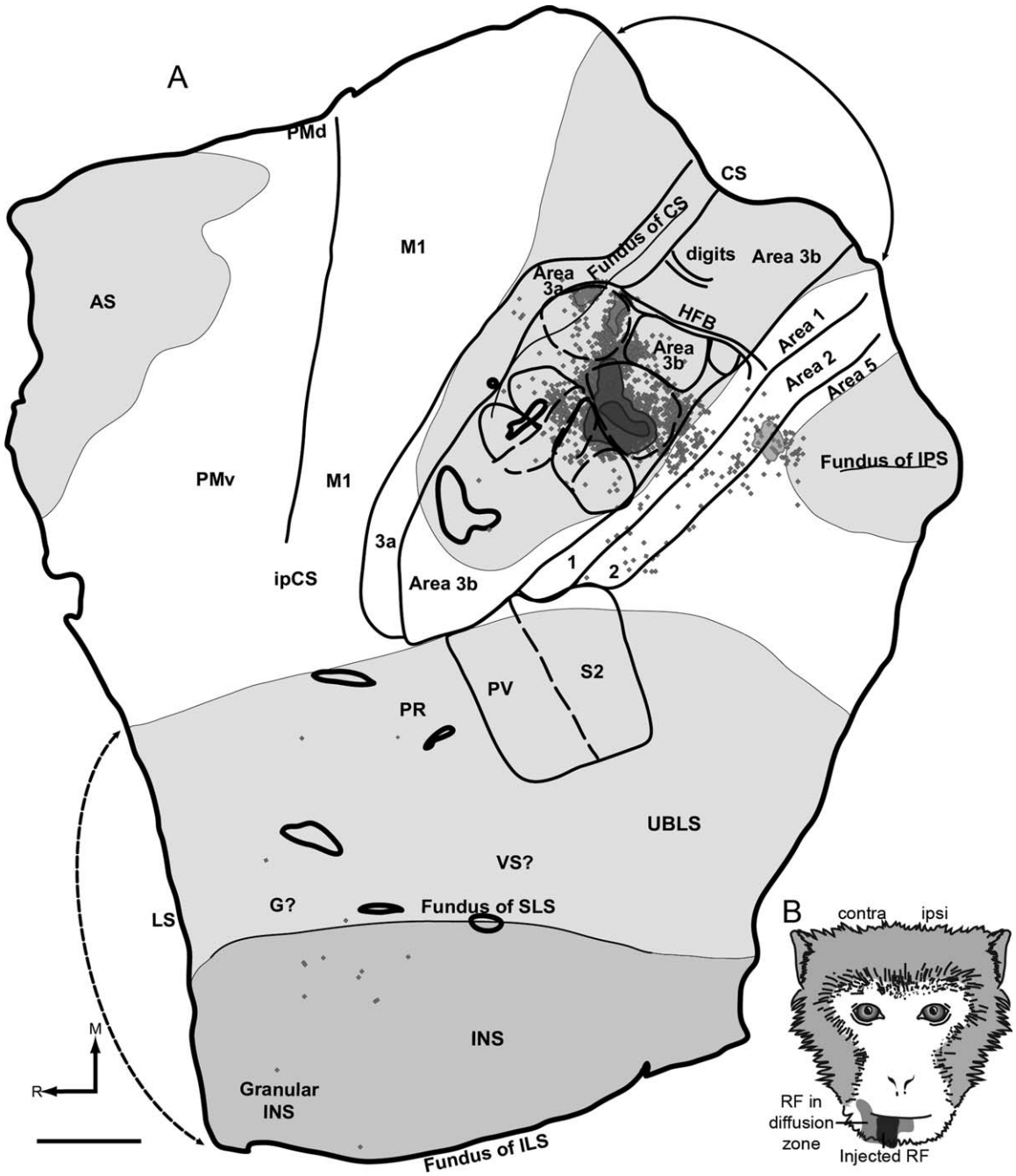


Figure 9. Case 2 BDA injection into 3b lip/chin. **A:** Distribution of BDA-labeled cells through the somatosensory cortex of case 2. Each gray diamond is one BDA-filled cell. The injection of BDA was centered on the representation of the medial lower lip and anterior chin but spread to include representations of most of the lower lip and the rest of the chin. As with other case 2 injections, this one labeled cells in a pattern similar to injections in other parts of 3b, with strong local inputs coming from within the injected representation, more inputs from other nearby representations within 3b, and then inputs from other somatosensory areas. Although this was our most medial injection, no BDA back-filled cells were found medial to the hand–face border. For abbreviations see list. Conventions as in Figure 6. **B:** Injected receptive field. Scale bar = 5 mm.

lateral sulcus, perhaps because of poor tracer transport. The few scattered neurons labeled in the insular cortex might have been the result of the BDA injection spread into the tongue region.

Case 2: lip representation

A small injection of FR was placed into the representation of the contralateral middle and medial lower lip (Fig. 10) between cores of the tongue/lip (CTB) and

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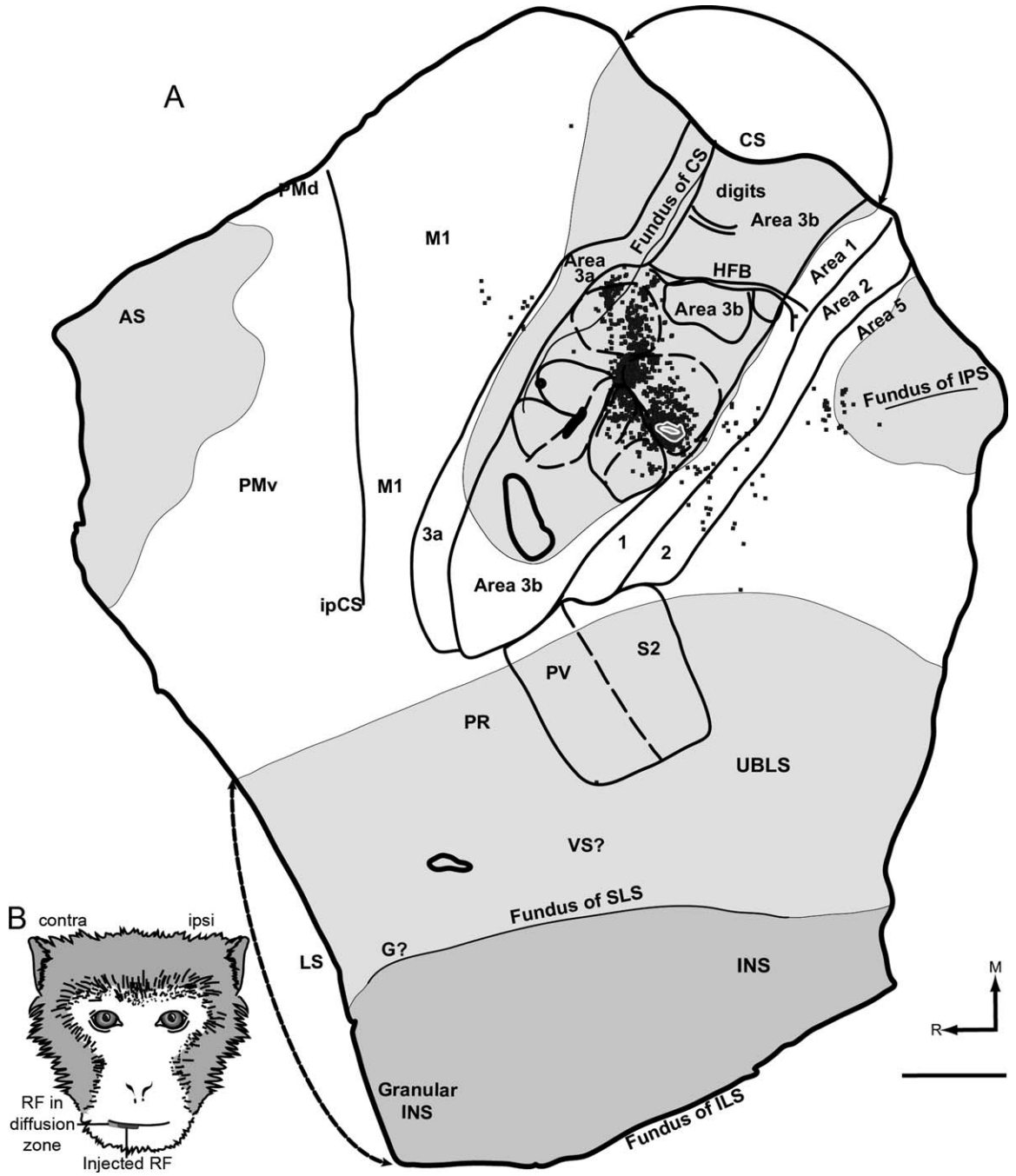


Figure 10. Case 2 FR injection into 3b lip. **A:** Distribution of FR-labeled cells through the somatosensory cortex of case 2. The injection of FR centered on the lower lip representation was small but does show a pattern of cells projecting to the 3b lip region from within the 3b face responsive cortex, areas 3a, 1, 2, and M1, that is similar to that revealed by injections into the tongue and lip representations. For abbreviations see list. Conventions as in Figure 6. **B:** Injected receptive field. Scale bar = 5 mm.

lip/chin (BDA) injections in case 2 (Fig. 5B). This FR injection included representations of the skin and hair along the rest of the contralateral lower lip. A region of cortex with weak responses to the contralateral lower incisors might also have been involved in the FR injection, but these penetrations were on the edge of the injection site, and it is unclear how much tracer uptake occurred in this representation of the teeth.

The 3b lower lip FR injection was small, but it produced a pattern of labeled neurons that was similar to that revealed by the other case 2 injections (Fig. 10). Back-filled neurons were primarily in the representations

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of the lower lip and chin. Again, the presumptive upper face representation in the unmapped part of 3b just lateral to the hand–face border contained labeled neurons. The most medial portion of cortex representing the tongue also provided inputs to the injected lip region. No label was found in 3b hand cortex (Table 2). Areas 3a, 1, 2, and 5 also provided inputs to the lip representation. Projections from motor and lateral sulcus somatosensory cortex were also indicated by FR-filled cells. Unlike the CTB and BDA injections in case 2, no region of cortex responding to the tongue was involved in the FR injection, and no neurons back-filled with FR were found in presumptive taste areas of the upper bank of the lateral sulcus and insula.

Case 3: tongue representation

CTB was injected in the CS into a region representing much of the middle and anterior tongue (Fig. 5C), centered on a region of responses to touch on the contralateral anterior and tip of the tongue (Fig. 11). Part of the contralateral middle palate was also included in the case 3 CTB injection. Our interpretation of the results of this CTB injection is complicated, somewhat, by a second injection of CTB into the area 5 hand representation in the contralateral hemisphere for an unrelated study. However, few if any of the labeled neurons in somatosensory regions representing the face would have been labeled by this second injection.

In case 3, CTB back-filled cells were distributed across much of nearby somatosensory cortex (Fig. 11). The locations of the lightly colored medial patch of filled neurons in more medial areas 1, 2, and 5 in Figure 10 are inconsistent with transport from the area 3b tongue injection and are attributed to the contralateral CTB injection. CTB-labeled cells in other, less-well-mapped somatosensory areas of the more caudal part of the upper bank of the lateral sulcus are shown in full and were counted when we determined magnitudes of projections to ensure that we did not miss any cells projecting to the 3b tongue representation (Table 1). The results revealed strong projections to the 3b tongue region from the area 3b mouth (the rest of the tongue, teeth, and palate) and face (chin, lips, and weaker from the face) representations contained labeled cells (Fig. 11). As with the other cases with tongue injections, almost no labeled cells were seen in 3b hand cortex (Table 2). Again, there were projections to the 3b tongue representation from areas 3a, 1, 2, S2, PV and presumptive VS and PR. Labeled cells indicating inputs from motor and premotor cortex were concentrated laterally in M1 and PMv. Finally, there were CTB-filled neurons near the superior limiting sulcus at the extreme anterior junction of the upper bank of the lateral sulcus and insula, suggesting projections from gustatory cortex to the 3b tongue representation.

Case 3: lateral representation of the teeth

The WGA-HRP injection in case 3 was a large injection in the most lateral portion of 3b on the surface of the brain beyond the lateral tip of the central sulcus (Fig. 5C). Although centered on a representation of the contralateral upper first incisor, this WGA-HRP injection was placed into a region containing responses to all of the anterior teeth as well as parts of the gums on both sides of the mouth, including the bilateral upper and lower anterior teeth and surrounding gum, the contralateral lower first premolar, the ipsilateral upper first premolar, all of the contralateral hard palate, and the bilateral anterior hard palate (Fig. 12). Although we believe that the zone of uptake from this WGA-HRP injection was restricted, the diffusion zone around the injection core was large and included representations of the ipsilateral upper second incisor and canine; the bilateral lower incisors, canines, and premolars; and the bilateral anterior half of the tongue. It also spread into unmapped regions of areas 1, PV, and S2 that likely also represented the face (Fig. 5C).

Much of the ipsilateral cortex contained labeled neurons (Fig. 12, Table 1). There were strong inputs from the rest of the teeth, tongue, palate, lip, and face representations within area 3b. Although the filled neurons outside of area 3b revealed by the case 3 WGA-HRP injection cover a wider region of cortex than the injection into the medial representation of the teeth in case 1, the pattern of these projections was similar to both the case 1 FR injection in the medial region of the teeth and that revealed by injections into the 3b tongue cortex in all cases. As with the medial representation of the teeth, this injection centered on the lateral representation of the teeth failed to label cells in the region of the anterior upper bank of the lateral sulcus and anterior insula that contained filled neurons after case 3's tongue injection.

Case 4: tongue representation

The case 4 injection of CTB into the tongue representation was centered on a region responding to stimulation of the contralateral middle and anterior tongue without the tip (Fig. 13). This injection of CTB spread to include representations of the rest of the contralateral tongue, the anterior half of the ipsilateral tongue, and the tip of tongue; the gums around the contralateral lower premolars and first molar; the upper first and second molars and the ipsilateral upper canine and premolars; and all of the anterior palate and the contralateral half of the rest of the palate. The diffusion zone included an adjacent part of area 1 (Fig. 5D).

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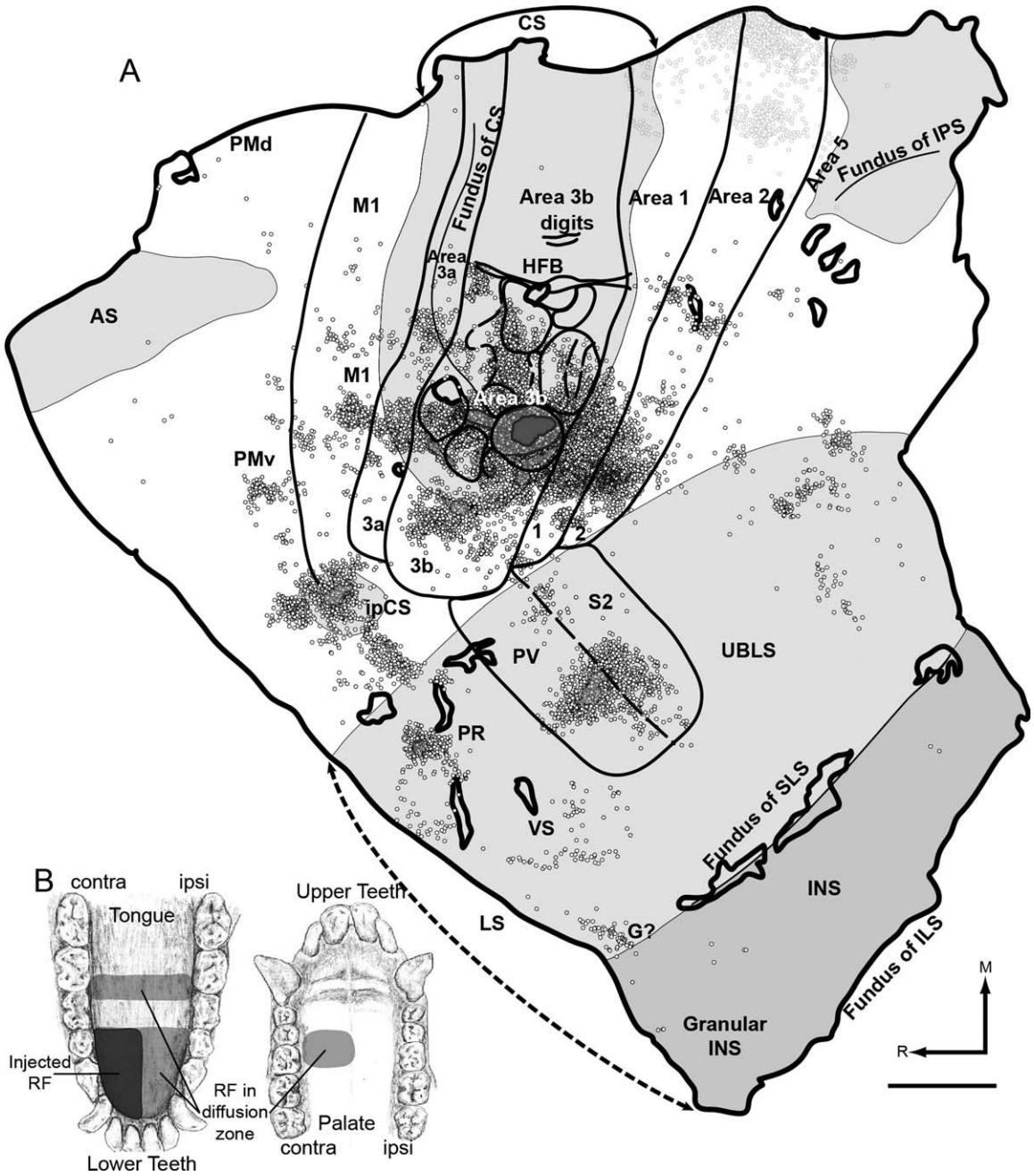


Figure 11. Case 3 CTB injection into 3b tongue. **A:** Distribution of CTB labeled cells through the somatosensory cortex of case 3. In case 3, CTB was injected into a region including responses to the tongue and part of the contralateral palate. Projections to this tongue region were largely similar to those of other cases, with strong local connections, projections from other somatosensory and motor areas, and input coming from presumptive taste cortex. The medial patch of lightly shaded cells likely results from a contralateral injection placed for another study. For abbreviations see list. Conventions as in Figure 6. **B:** Injected receptive field. Scale bar = 5 mm.

Resulting CTB label revealed strong projections from other parts of the oral cavity in 3b as well as anterior parietal cortex, lateral sulcus cortex, insular cortex, and motor cortex (Fig. 13, Table 1). Intraoral representations containing CTB-filled neurons after the injection into the 3b tongue region included other parts of the tongue representation and representations of the palate, gingiva, teeth, and buccal wall. The inputs from the representations of the teeth and gums came only from the more lateral representations, not from the more medial part of cortex that responded to stimulation of the teeth and gums. The region of mixed lower lip and chin responses in 3b projected to the region of the 3b tongue representation injection. Fewer labeled cells

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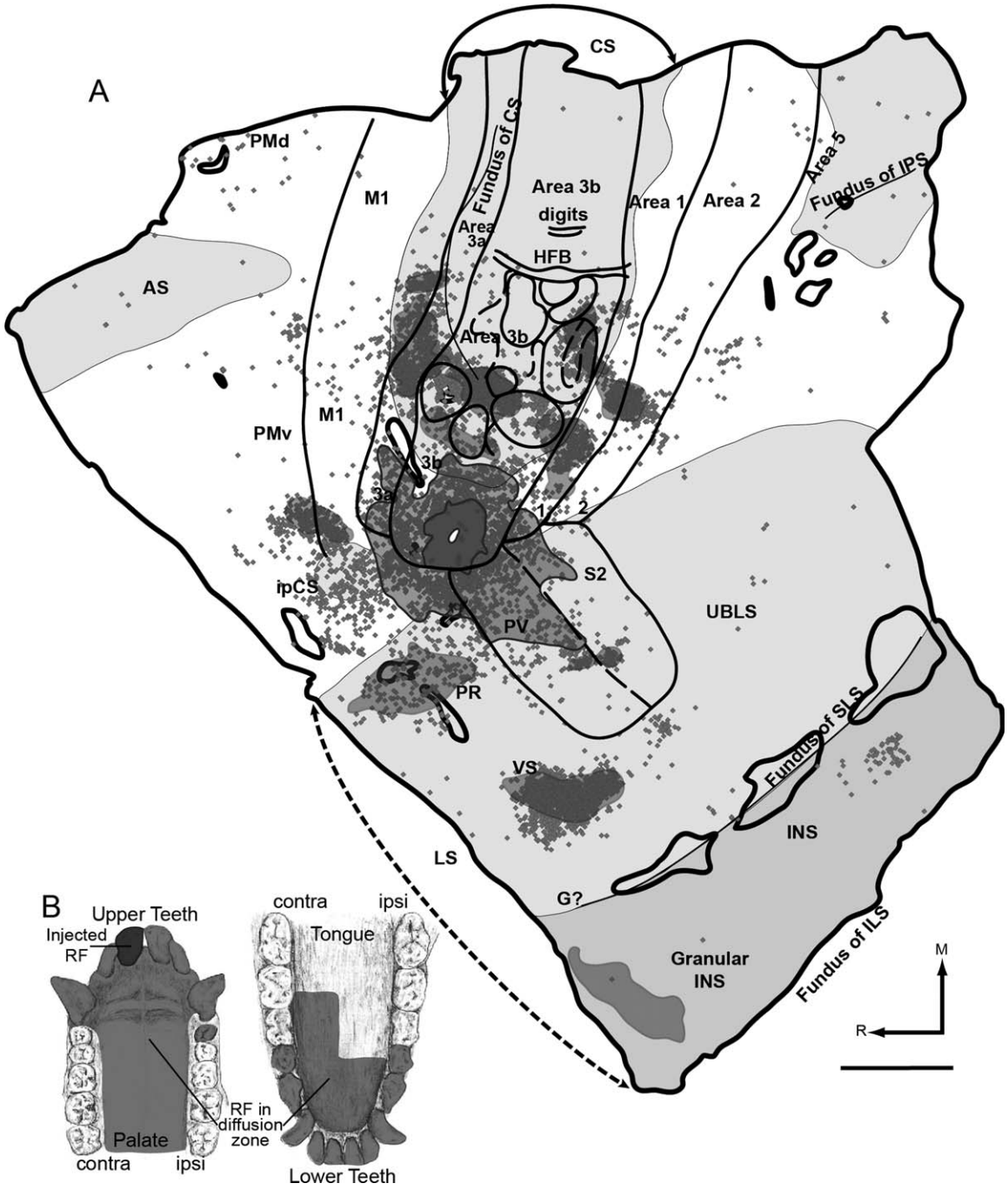


Figure 12. Case 3 WGA-HRP injection into lateral 3b teeth. **A:** Distribution of WGA-HRP-labeled cells through the somatosensory cortex of case 3. Each gray diamond is one WGA-HRP-filled cell. WGA-HRP injected into the representation of the teeth, gums, and palate in area 3b resulted in a pattern of inputs similar to that of the medial teeth representation injection in case 1. Label in the anterior upper bank of the lateral sulcus and anterior insula after this injection was in contrast to that of the case 1 teeth representation injection; however the diffusion zone of this injection did include parts of the tongue representation in 3b and possibly other areas. For abbreviations see list. Conventions as in Figure 6. **B:** Injected receptive field. Scale bar = 5 mm.

were found in the upper lip and face representations in 3b. Almost no CTB-filled neurons were found in the 3b hand cortex (Table 2). In addition to these local connections, there were projections from all of the anterior parietal and lateral sulcus somatosensory areas, and lateral M1 and PMv to the 3b tongue representation in case 4. Several filled neurons were found in the anterior upper bank of the lateral sulcus and insula. These inputs indicate that area G and other taste regions may project to area 3b.

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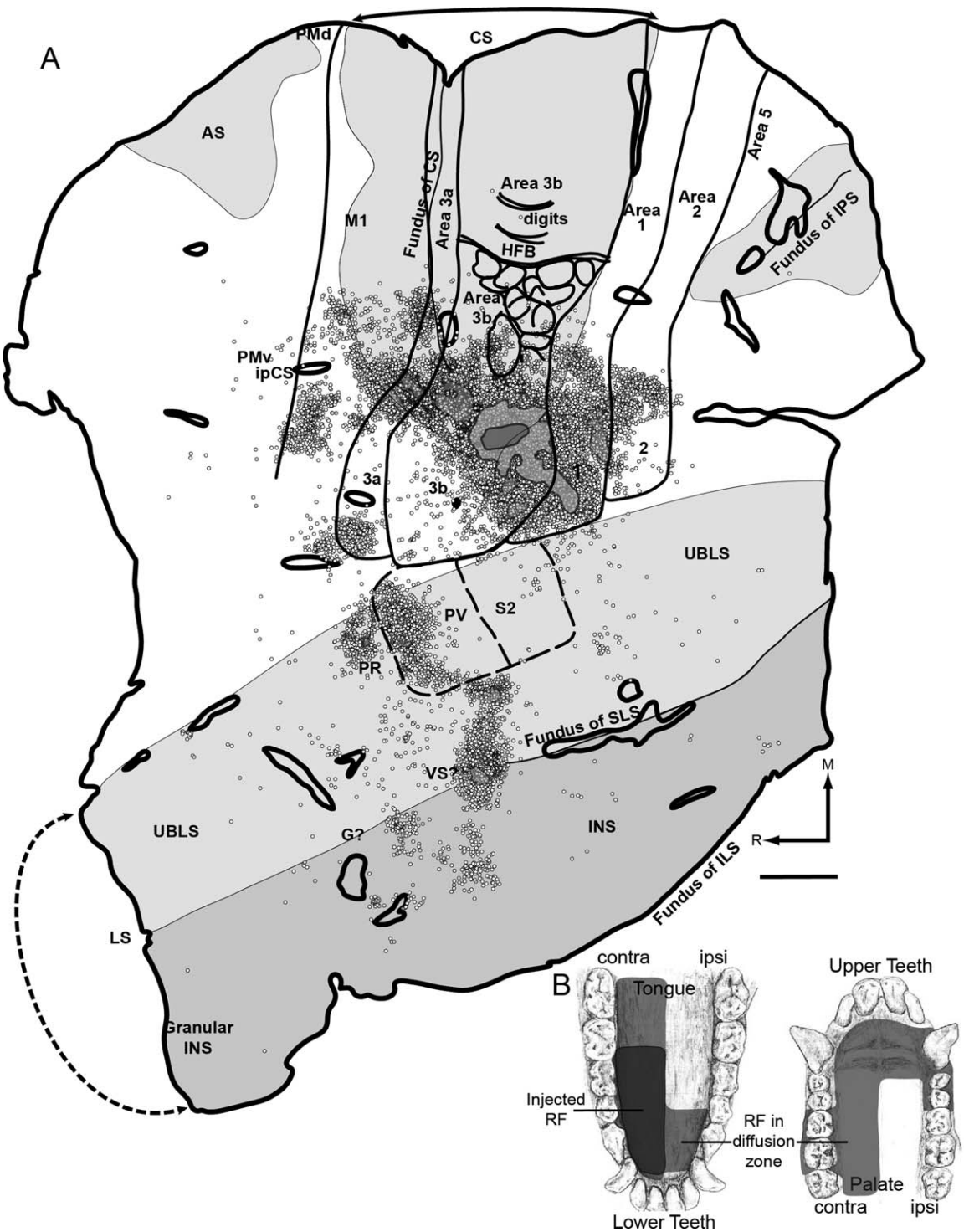


Figure 13. Case 4 CTB injection into 3b tongue. **A:** Distribution of CTB-labeled cells through the somatosensory cortex of case 4. A CTB injection into the tongue region in case 4 reveled a pattern of strong local connections, projections from somatosensory, motor, and possible taste cortex similar to those in other tongue injections. For abbreviations see list. Conventions as in Figure 6. **B:** Injected receptive field. Scale bar = 5 mm.

Case 4: lateral representation of the teeth

Representations of the ipsilateral upper incisors and canine and the ipsilateral lower second incisor, canine, and first premolar were at the center of case 4's WGA-HRP injection in rostrolateral 3b on the surface of the brain (Figs. 5D, 13). The injected region also included cortex with responses to stimulation on the rest of the ipsilateral upper teeth; the ipsilateral lower first molar,

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second premolar, and first incisor; and the contralateral lower anterior teeth. The representation of the gums surrounding the ipsilateral upper anterior teeth was also a part of this WGA-HRP injection.

As for the injections in our other cases, the case 4 WGA-HRP injection into the lateral representation of the teeth resulted in many back-filled cells in other parts of the mouth representation in area 3b (Fig. 14, Table 1). There was a high concentration of labeled neurons within the lateral teeth/gum representation, but none was found in the more medial representation of the teeth. Regions processing signals from the palate, cheek pouch, gums, and tongue also projected to the lateral representation of the teeth. Although both the lower and the upper lip representations contained labeled neurons, the connections coming from the lower lip appeared to be denser. Almost all of the filled cells in area 3b were in the representations of the mouth and face (Table 2). Outside of primary somatosensory cortex, there were labeled cells indicating projections from other somatosensory areas in the anterior parietal cortex, including 3a, 1, 2, and 5, S2, PV, and presumptive PR and VS. Filled cells were found in lateral M1 and PMv, and the field of labeled cells spread onto the caudal bank of the arcuate sulcus. There were fewer labeled neurons in the anterior part of the upper bank of the lateral sulcus and anterior insula than with the case 4 tongue representation injection.

Contralateral inputs to the 3b tongue representation

Contralateral inputs to area 3b were analyzed only for the CTB injections into cortical representations of the tongue in each case. Our injections into the 3b tongue representation consistently resulted in labeled neurons in the contralateral hemisphere in all of our cases (Fig. 15). Based on their location in relation to the myeloarchitecture, CTB-filled cells were found in the matching 3b tongue representation of the contralateral hemisphere. This indicates that the lateral part of area 3b in one hemisphere projects to the tongue representation on the other side of the brain. In addition, many labeled neurons were anterior to area 3b in area 3a and primary motor cortex. Small numbers of projections from contralateral areas 1 and 2 and somatosensory areas in the upper bank of the lateral sulcus were also indicated.

DISCUSSION

Our results appear to be the first to reveal the cortical network for the processing of tactile information from the inside of the mouth in macaque monkeys. Injections of anatomical tracers into 3b representations of the tongue and teeth revealed projections to this region from a number of nearby regions cortex (Fig. 16). Oral cavity representations had strong intrinsic connections with the rest of the 3b mouth and face regions. Outside of area 3b, inputs to the mouth and face regions of area 3b arose from areas 3a, 1, 2, and 5; S2; PV; PR and VS areas; M1; and PMv. Cortex in the anterior part of the upper bank of the lateral sulcus and insula and around the principal sulcus contained only neurons filled by injections into the representation of the tongue in area 3b. Injections into representations of the lips and chin in area 3b labeled neurons largely in the rest of face and oral cavity cortex, with little to no input from other body part representations in area 3b, including that of the hand.

Network for processing somatosensory information from the oral cavity

The neural network for processing somatosensory information from the inside of the mouth and the face is large and complex and is similar to that for the hand. All somatosensory areas in the anterior parietal cortex (3b, 3a, 1, and 2) and the lateral sulcus (S2, PV, PR, and PV) as well as associated areas in the PPC, contained filled cells after our injections into 3b face and mouth representations (Fig. 16, solid arrows). Additionally, motor and premotor areas also provided input to the representations of intraoral structures and the face in area 3b. Similar patterns of inputs to the 3b oral cavity and face representations have been shown in New World owl, squirrel, and marmoset monkeys (Iyengar et al., 2007). Tracer injections into representations of other body parts in area 3b have demonstrated a similar pattern of projections across all primates (Pons and Kaas, 1985, 1986; Kaas and Pons, 1988; Krubitzer and Kaas, 1990; Darian-Smith et al., 1993; Burton and Fabri, 1995; Qi et al., 2002; Wu and Kaas, 2003; Coq et al., 2004; Kaas, 2004). Thus, our results reveal a network for processing tactile information from the inside of the mouth that is similar to that for other body parts and likely is common to all primates. Nevertheless, greater numbers of neurons are involved in this network as a result of the great amount of cortex representing the face and mouth.

Because each of our injections was placed into an electrophysiologically defined representation of the tongue, teeth, lip, or chin, the locations of labeled cells helped to identify poorly studied oral cavity and face representations in other somatosensory areas. Back-filled neurons were concentrated in the lateral parts of areas 3a, 1, and 2. We did not map these regions

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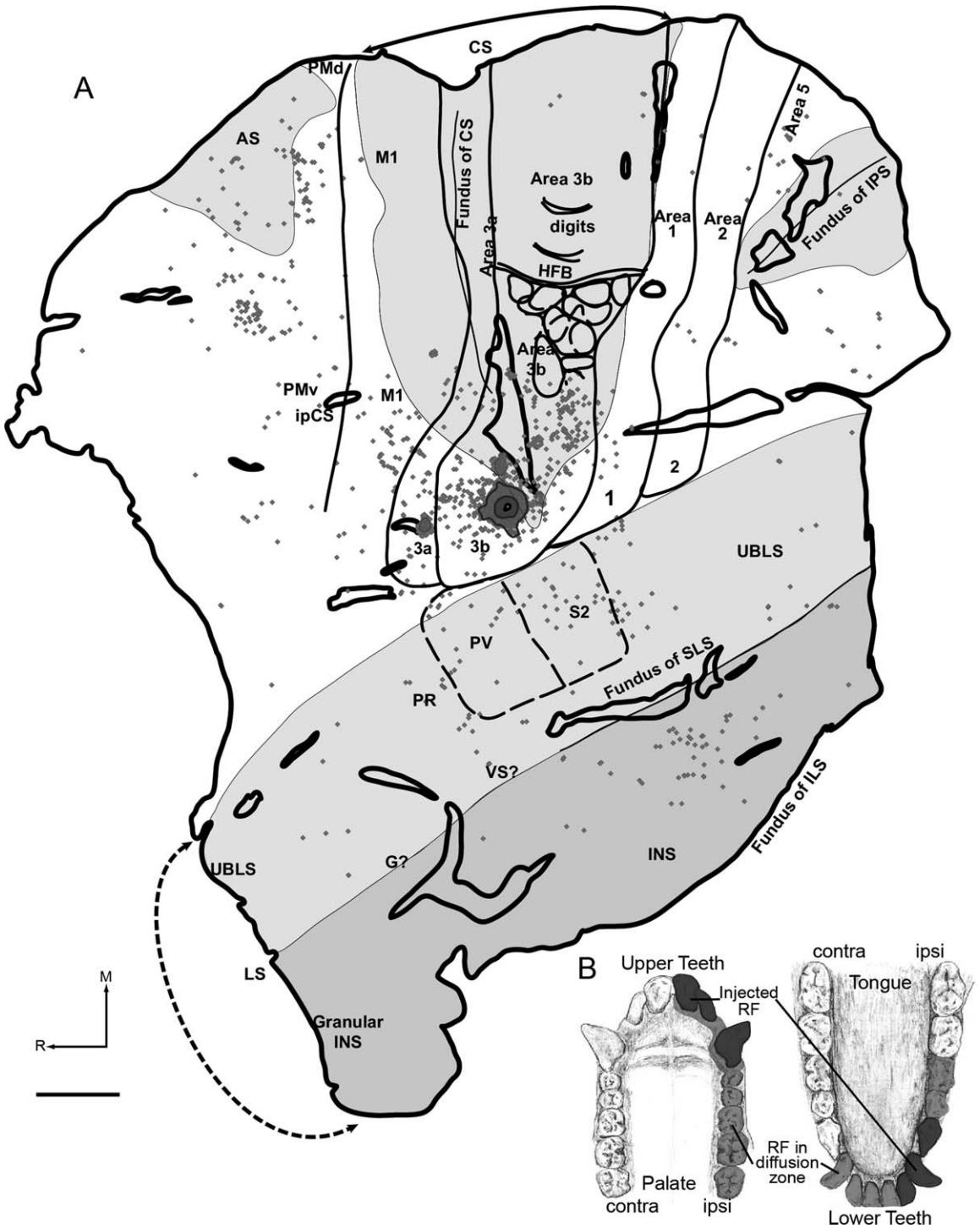


Figure 14. Case 4 WGA-HRP injection into lateral 3b teeth. **A:** Distribution of WGA-HRP-labeled cells through the somatosensory cortex of case 4. In case 4, WGA-HRP was injected into a representation mostly of the ipsilateral teeth. Local projections to this region were strong. Inputs from areas 3a, 1, 2, S2/PV, M1, and PMv were similar to those of other cases. As with the other teeth representation injections, few labeled cells were found in presumptive taste cortex. For abbreviations see list. Conventions as in Figure 6. **B:** Injected receptive field. Scale bar = 5 mm.

during our experiments, but limited mapping in these regions in macaques and New World monkeys suggests that representations of the face and mouth lie in the lateral region of each of these areas (Nelson and Kaas, 1981; Pons et al., 1985a,b; Manger et al., 1995, 1996, 1997; Huffman and Krubitzer, 2001; Jain et al., 2001a; Qi and Kaas, 2004; Iyengar et al., 2007; Seelke et al., 2012). Given that the representations of any body part

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in each of the anterior parietal areas are preferentially connected to the matching body part representation in area 3b (Burton and Fabri, 1995), it is likely that the projections from areas 3a, 1, and 2 revealed by our injections identify the oral cavity and face representations in each of these areas. Face and mouth representations have not been thoroughly described in area 5. However, lesions in the face region of the primary somatosensory face cortex resulted in degeneration in lateral areas 5–7 (Pearson and Powell, 1978, 1985). Our injections of retrograde tracers resulted in labeled cells in the cortex just caudal to the area 2 face representation (Pons et al., 1985a,b), a lateral portion of area 5–7 likely to represent the face and mouth (Padberg et al., 2009). Previously, projections from area 5 to area 3b were described as weak, although they originated from same approximate mediolateral level in area 5 as the injection in area 3b (Burton and Fabri, 1995), likely matching body part representations.

There are several proposed somatosensory areas in the cortex of the lateral sulcus, and most or all of the regions of these areas contained labeled neurons after our injections. Although electrophysiological mapping of these areas has been limited by their position in the lateral sulcus, the somatotopic organizations of S2, PV, and parts of VS have been described in macaques (Krubitzer et al., 1995). The representation of the oral cavity was not completely mapped in S2 and PV, but the locations responsive to stimulation of the face and intraoral structures were often nonadjacent. The precise organization of the map of the face and mouth in these regions of S2 and PV is unclear, but the recorded responses to stimulation on the face and mouth were from neurons concentrated near the crest of the lateral sulcus closer to the junction of S2 and PV with areas 3b, 1, and 2. Likewise, labeled cells in S2 and PV tended to be concentrated dorsally, although the distributions of labeled neurons often extended toward the ventral border, deeper in the lateral sulcus. Receptive fields for neurons in S2 and PV are often large (Krubitzer et al., 1995), leaving the organizations of the maps in these areas not as precise as that in area 3b, where smaller receptive fields dominate. Thus, it may be that representations of larger regions of the skin surface of the mouth and face in area S2 and PV project to the

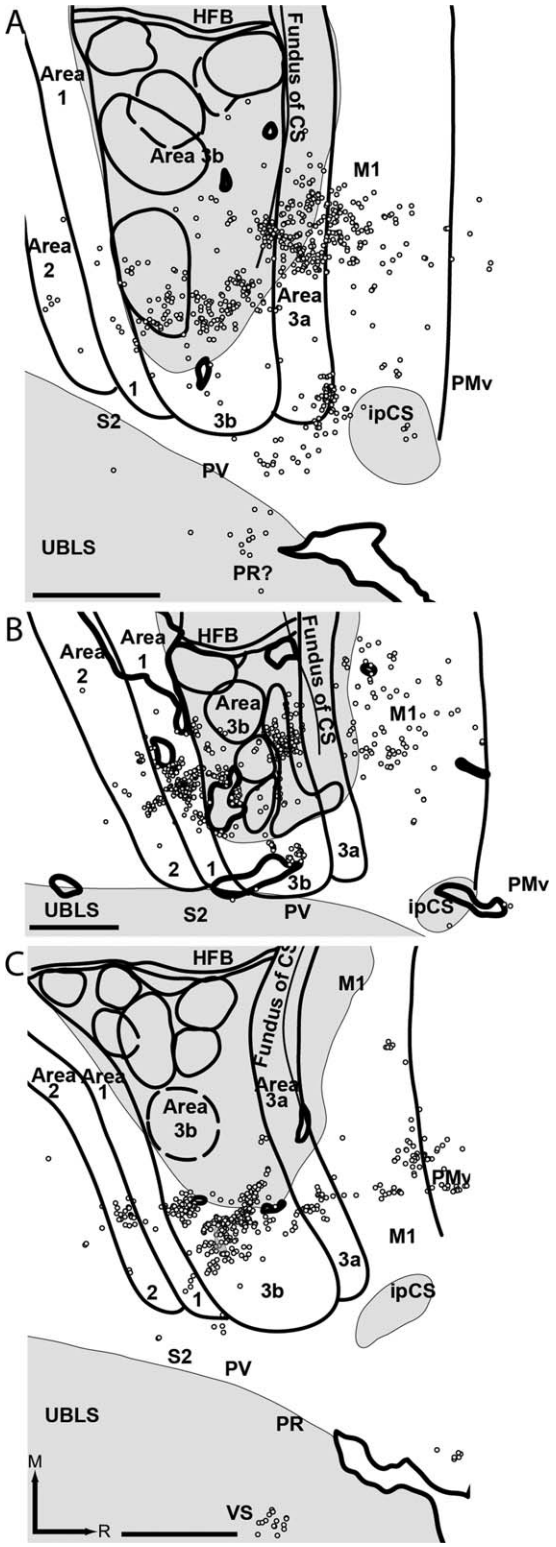


Figure 15.

Figure 15. Contralateral projections to the 3b tongue representation. **A–C:** Distribution of CTB labeled cells through the contralateral somatosensory and motor cortex of cases 1 (A), 2 (B), and 4 (C). In all cases, cells labeled after CTB injections into the area 3b tongue representation were concentrated in the lateral part of the contralateral are 3b, presumably the matching tongue representation. Back-filled cells were also found in contralateral area 3a, 1, 2, M1, and PMv in all cases. A few projections from other somatosensory areas (S2, PV, PR, and VS) were variably indicated. For abbreviations see list. Rostral is right, medial is up, all other conventions as in Figure 6.

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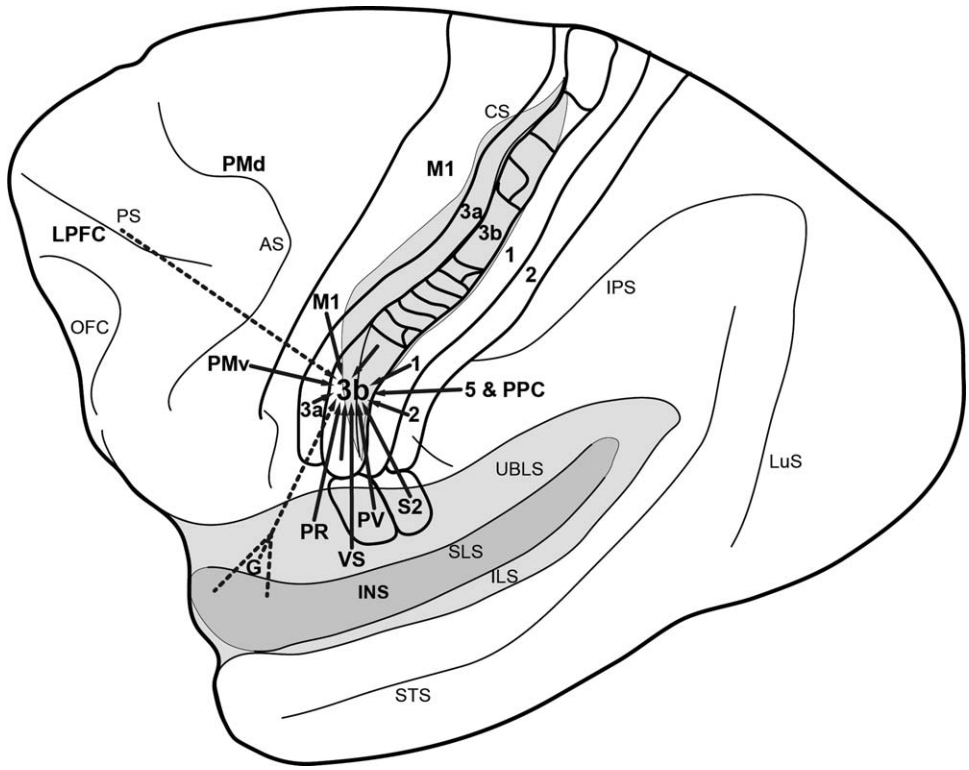


Figure 16. Summary of projections to the 3b representation of the oral cavity and face. The 3b representations of the oral cavity and face receive inputs from within this region, the presumptive mouth and face representations in the other somatosensory areas of the anterior and posterior parietal cortex and in the lateral sulcus, and regions of motor and premotor cortex involved in controlling movements of the mouth and face (solid lines). Presumptive area G and regions of the insula implicated in processing taste as well as cortex in the dorsal bank of the principal sulcus were shown to project only to the 3b tongue representation (dashed lines). For abbreviations see list.

more circumscribed representations of individual oral cavity and facial structures in area 3b. Organized projections from S2 and PV to area 3b have been reported for other body parts in New World monkeys and prosimian galagos, in which S2 and PV are more accessible for study, and physiological maps are more complete (Krubitzer and Kaas, 1990; Qi et al., 2002; Wu and Kaas, 2003; Coq et al., 2004; Iyengar et al., 2007). The existence of a distinct representation of the inside of the mouth in areas PR and VS has been demonstrated only by limited mapping of responses from stimulation of the mouth and face (Krubitzer et al., 1995; Qi et al., 2002; Coq et al., 2004) and the results of previous tracer studies in New World monkeys (Iyengar et al., 2007). The present evidence for projections to face and mouth representations in macaques provides further support for the existence of face and oral cavity representations in PR and VS, although more extensive physiological explorations of these areas will be required to identify face and mouth representations with confidence.

Our injections into the 3b representations of the oral cavity and face also labeled neurons in lateral M1 and PMv. Movements of the face and mouth have been elicited by stimulation of these regions in both New World monkeys and macaques (McGuinness et al., 1980; Ghosh and Gattera, 1995; Graziano et al., 2002; Stepniewska et al., 2006; Gharbawie et al., 2011). Furthermore, cells in this region have been shown to be active during movements of the jaw and tongue (Hoffman and Luschei, 1980; Murray and Sessle, 1992). In addition, lateral positions of both somatosensory and motor cortex must be functioning properly for a monkey to perform a tongue protrusion task properly (Murray et al., 1991; Lin et al., 1993). The homotopic projections from M1 and PMv to area 3b representations of structures in the mouth and face that we have demonstrated here likely reflect parts of a network for the proper integration of sensory information to guide movements of the face and mouth during feeding and vocalization.

Projections unique to the 3b tongue representation

Although most of the projections to the 3b representations of the face and mouth were similar to those

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demonstrated for other body part representations in area 3b, the representation of the tongue appears to receive unique projections from several areas implicated in the cortical system for processing taste (Fig. 16, dashed arrows). Primarily, these inputs arose from cortex of the anterior upper bank of the lateral sulcus and insula. The cortical system for processing taste is not fully understood. There is general agreement about the existence of a primary gustatory area, G, in macaques (Ogawa et al., 1985, 1989; Ogawa, 1994; Cipolloni and Pandya, 1999; Scott and Plata-Salamán, 1999) that is similar in location to that described by Sanides (1968) for squirrel monkeys. G is often defined anatomically as a region of cortex in the upper bank of the lateral sulcus that receives strong input from the ventroposterior medial parvicellular nucleus (VPMpc), the taste nucleus of the thalamus (Pritchard et al., 1986). Functionally, this area is difficult to define as a result of sparse responses to taste stimuli (Scott et al., 1986; Ogawa et al., 1989; Plata-Salamán and Scott, 1992; Pritchard and Norgren, 2004) and fairly high incidence of cells that respond to touch (Scott et al., 1986; Ogawa et al., 1989). Nearby regions of the insula have also been implicated in the processing of taste (Ogawa et al., 1989; Plata-Salamán and Scott, 1992; Kringelbach, 2004; Pritchard and Norgren, 2004). It is likely that the cells back-filled by our injections into the tongue representations in our cases were in area G and possibly these other gustatory related areas. In macaques, there is no projection from VPMpc to the 3b tongue representation (Cerkevich et al., 2009, 2013). Thus, it appears that projections from taste-related cortex to the macaque 3b tongue representation form the only pathway by which gustatory information can directly access the primary sensory cortex for the tongue. However, we cannot rule out the possibility that it is only the somatosensory responsive cells in these gustatory regions that project to area 3b.

Both the orbitofrontal cortex (OFC) and the lateral prefrontal cortex (LPFC) have been implicated in the qualitative evaluation of taste stimuli (Rolls et al., 1989; Rolls and Baylis, 1994; Carmichael and Price, 1995; Rolls, 2000; Kringelbach, 2004; Kringelbach et al., 2004) and may be vital to integrating information from multiple sensory systems to create the rewarding response often associated with food (Shepherd, 2012). Our injections did not label cells in the OFC, but a few neurons in the LPFC were labeled by our injections into the tongue region. Connections between the LPFC and area 3b have not been previously demonstrated (Yeterian et al., 2012), and the projection we found was very weak, consisting of only eight of the 18,447 total CTB cells labeled by a 3b tongue representation injection in case 1. However, in macaques, cells in the LPFC have been shown to change activity during delay periods in tasks in which food rewards were visible or associated but not seen (Watanabe, 1996; Hikosaka and Watanabe, 2000). Responses to taste (Kringelbach et al., 2004) and changes in fatty acid concentrations in the blood after a meal (Tataranni et al., 1999) have been demonstrated in the dorsolateral prefrontal cortex in humans. These responses in humans appear to be in regions of cortex that correspond the location of our labeled cells in LPFC (Petrides et al., 2012).

Face–hand interaction in area 3b

It has long been known that sensory loss from the hand produced by peripheral amputations (Merzenich et al., 1984; Florence and Kaas, 1995) and deafferenting injuries of the peripheral nerves (Merzenich et al., 1983a,b; Garraghty and Kaas, 1991; Garraghty et al., 1994) and spinal cord (Pons et al., 1991; Jain et al., 1997, 1998b, 2008) can lead to changes in the responsiveness of the region of area 3b representing the hand. Reorganization in the somatosensory cortex after such injuries often results in hand cortex becoming responsive to stimulation on the face (Merzenich et al., 1984; Pons et al., 1991; Florence and Kaas, 1995; Jain et al., 1997, 1998b, 2008; Manger et al., 1997). When small injections of distinguishable fluorescent tracers were placed into mapped representations of the 3b hand and face in fully intact owl, squirrel, and marmoset monkeys, almost no connections between the two regions of area 3b were found (Fang et al., 2002). Intrinsic connections of the 3b hand have been shown to be confined to a mediolaterally restricted region of area 3b (Lund et al., 1993; Burton and Fabri, 1995). Although the full extent of the hand representation was not mapped and the hand–face border was not defined in these studies, few of these connections appeared to extend into the representation of the face that lies laterally in 3b compared with the hand representation. Our injections into representations of the face and intraoral structures resulted in very few, if any, labeled cells in area 3b medial to the hand face border (Table 2). These combined results suggest there is no preexisting face–hand network in area 3b. Thus, the reorganization that occurs after deafferenting injuries must result from changes in the intrinsic connections of area 3b that emerge after sensory loss or changes in the connections of subcortical levels of somatosensory processing (Florence and Kaas, 1995; Florence et al., 2000).

CONCLUSIONS

By injecting anatomical tracers into electrophysiologically defined representations of the tongue, teeth, and

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face in the primary somatosensory cortex, we were able to reveal the projections to the 3b oral cavity region in macaques. This is the first time that the network for somatosensory processing from the oral cavity has been described for Old World monkeys. For the most part, the projections to area 3b's oral cavity representation from other somatosensory and motor areas are similar to those going to the better studied 3b hand representation. The tongue representation, however, received idiosyncratic inputs from taste-related areas buried deep in the lateral sulcus and insula and very weakly from the lateral prefrontal cortex. Finally, no network of intrinsic connections in area 3b exists in normal monkeys that would support the reactivation of area 3b hand cortex by stimulation the face after long-standing sensory loss form the hand.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to declare.

ROLE OF AUTHORS

All authors had full access to all the data in the study and take responsibility from the integrity of the data and the accuracy of the data analysis. Study concept and design: JHK, H-XQ Acquisition of data: CMC, H-XQ. Analysis and interpretation of data: CMC, JHK, H-XQ. Drafting of the manuscript: CMC, JHK, H-XQ. Obtained funding: JHK.

LITERATURE CITED

Angelucci A, Clasca F, Sur M. 1996. Anterograde axonal tracing with the subunit B of cholera toxin: a highly sensitive immunohistochemical protocol for revealing fine axonal morphology in adult and neonatal brains. J Neurosci Methods 65:101–112.

Avivi-Arber L, Martin R, Lee J-C, Sessle B. 2011. Face sensorimotor cortex and its neuroplasticity related to orofacial sensorimotor functions. Arch Oral Biol 56:1440–1465.

Bruce K, Grofova I. 1992. Notes on a light and electron microscopic double-labeling method combining anterograde tracing with *Phaseolus vulgaris* leucoagglutinin and retrograde tracing with cholera toxin subunit B. J Neurosci Methods 45:23–33.

Burton H, Fabri M. 1995. Ipsilateral intracortical connections of physiologically defined cutaneous representations in areas 3b and 1 of macaque monkeys: projections in the vicinity of the central sulcus. J Comp Neurol 355:508–538.

Carmichael ST, Price JL. 1995. Sensory and premotor connections of the orbital and medial prefrontal cortex of macaque monkeys. J Comp Neurol 363:642–664.

Cerkevich C, Qi H-X, Kaas J. 2009. Thalamocortical connections of the macaque 3b oral cavity representation. Program No. 464.12. Neuroscience 2009 Abstracts. Chicago, IL: Society for Neuroscience, 2009. Online.

Cerkevich C, Qi H-X, Kaas JH. 2013. Thalamic input to representations of the teeth, tongue, and face in somatosensory area 3b of macaque monkeys. J Comp Neurol (in press).

Cipolloni PB, Pandya DN. 1999. Cortical connections of the frontoparietal opercular areas in the rhesus monkey. J Comp Neurol 403:431–458.

Coq J-O, Qi H, Collins CE, Kaas JH. 2004. Anatomical and functional organization of somatosensory areas of the lateral fissure of the New World titi monkey (*Callicebus moloch*). J Comp Neurol 476:363–387.

Darian-Smith C, Darian-Smith I, Burman K, Ratcliffe N. 1993. Ipsilateral cortical projections to areas 3a, 3b, and 4 in the macaque monkey. J Comp Neurol 335:200–213.

Fang P-C, Jain N, Kaas J. 2002. Few intrinsic connections cross the hand-face border of area 3b of New World monkeys. J Comp Neurol 454:310–319.

Florence SL, Kaas JH. 1995. Large-scale reorganization at multiple levels of the somatosensory pathway follows therapeutic amputation of the hand in monkeys. J Neurosci 15:8083–8095.

Florence SL, Hackett TA, Strata F. 2000. Thalamic and cortical contributions to neural plasticity after limb amputation. J Neurophysiol 83:3154–3159.

Gallyas F. 1979. Silver staining of myelin by means of physical development. Neurol Res 1:203–209.

Garraghty PE, Kaas JH. 1991. Large-scale functional reorganization in adult monkey cortex after peripheral nerve injury. Proc Natl Acad Sci U S A 88:6976–6980.

Garraghty PE, Hanes DP, Florence SL, Kaas JH. 1994. Pattern of peripheral deafferentation predicts reorganizational limits in adult primate somatosensory cortex. Somatosens Mot Res 11:109–117.

Gharbawie OA, Stepniewska I, Kaas JH. 2011. Cortical connections of functional zones in posterior parietal cortex and frontal cortex motor regions in New World monkeys. Cereb Cortex 21:1981–2002.

Ghosh S, Gattera R. 1995. A comparison of the ipsilateral cortical projections to the dorsal and ventral subdivisions of the macaque premotor cortex. Somatosens Mot Res 12:359–378.

Graziano MSA, Taylor CSR, Moore T. 2002. Complex movements evoked by microstimulation of precentral cortex. Neuron 34:841–851.

Hikosaka K, Watanabe M. 2000. Delay activity of orbital and lateral prefrontal neurons of the monkey varying with different rewards. Cereb Cortex 10:263–271.

Hoffman DS, Luschei ES. 1980. Responses of monkey precentral cortical cells during a controlled jaw bite task. J Neurophysiol 44:333–348.

Huffman KJ, Krubitzer L. 2001. Area 3a: topographic organization and cortical connections in marmoset monkeys. Cereb Cortex 11:849–867.

Iyengar S, Qi H-X, Jain N, Kaas J. 2007. Cortical and thalamic connections of the representations of the teeth and tongue in somatosensory cortex of new world monkeys. J Comp Neurol 501:95–120.

Jain N, Catania K, Kaas J. 1997. Deactivation and reactivation of somatosensory cortex after dorsal spinal cord injury. Nature 386:495–498.

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Jain N, Catania K, Kaas J. 1998a. A histologically visible representation of the fingers and palm in primate area 3b and its immutability following long-term deafferentations. Cereb Cortex 8:227–236.

Jain N, Florence S, Kaas J. 1998b. Reorganization of somatosensory cortex after nerve and spinal cord injury. New Physiol Sci 13:143–149.

Jain N, Qi H-X, Catania K, Kaas J. 2001a. Anatomic correlates of the face and oral cavity representations in the somatosensory cortical area 3b of monkeys. J Comp Neurol 429:455–468.

Jain N, Qi H-X, Kaas J, Nicolelis M. 2001b. Long term chronic multichannel recordings from sensorimotor cortex and thalamus of primates. Nerw York: Elsevier. p 63–72.

Jain N, Qi H-X, Collins C, Kaas J. 2008. Large-scale reorganization in the somatosensory cortex and thalamus after sensory loss in macaque monkeys. J Neurosci 28:11042–11060.

Kaas J. 2004. Evolution of the large, complex sensorimotor systems of anthropoid primates. Int J Comp Psychol 17: 34052.

Kaas JH, Pons TP. 1988. The somatosensory system of primates. In: Steklis HD, Erwin JM, editors. Comparative primate biology, vol 4: neurosciences. New York: Alan R. Liss, Inc. p 421–468.

Kaas J, Qi H-X, Iyengar S. 2006. Cortical network for representing the teeth and tongue in primates. Anat Rec A Disco Mol Cell Evol Biol 288A:182–190.

Kringelbach ML. 2004. Food for thought: hedonic experience beyond homeostasis in the human brain. Neuroscience 126:807–819.

Kringelbach ML, de Araujo IET, Rolls ET. 2004. Taste-related activity in the human dorsolateral prefrontal cortex. Neuroimage 21:781–788.

Krubitzer LA, Kaas JH. 1990. The organization and connections of somatosensory cortex in marmosets. J Neurosci 10:952–974.

Krubitzer L, Clarey J, Tweedale R, Elston G, Calford M. 1995. A redefinition of somatosensory areas in the lateral sulcus of macaque monkeys. J Neurosci 15:3821–3839.

Lin LD, Murray GM, Sessle BJ. 1993. The effect of bilateral cold block of the primate face primary somatosensory cortex on the performance of trained tongue-protrusion task and biting tasks. J Neurophysiol 70:985–996.

Lin LD, Murray GM, Sessle BJ. 1994. Functional properties of single neurons in the primate face primary somatosensory cortex. I. Relations with trained orofacial motor behaviors. J Neurophysiol 71:2377–2390.

Lund JS, Yoshioka T, Levitt JB. 1993. Comparison of intrinsic connectivity in different areas of macaque monkey cerebral cortex. Cereb Cortex 3:148–162.

Manger PR, Woods TM, Jones EG. 1995. Representation of the face and intraoral structures in area 3b of the squirrel monkey (*Saimiri sciureus*) somatosensory cortex, with special reference to the ipsilateral representation. J Comp Neurol 362:597–607.

Manger PR, Woods TM, Jones EG. 1996. Representation of face and intraoral structures in area 3b of macaque monkey somatosensory cortex. J Comp Neurol 371:513–521.

Manger PR, Woods TM, Munoz A, Jones EG. 1997. Hand/face border as a limiting boundary in the body representation in monkey somatosensory cortex. J Neurosci 17:6338–6351.

McGuinness E, Sivertsen D, Allman JM. 1980. Organization of the face representation in macaque motor cortex. J Comp Neurol 193:591–608.

Merzenich M, Kaas J, Wall J, Sur M, Nelson R, Felleman D. 1983a. Progression of change following median nerve section in the cortical representation of the hand in areas 3b and 1 in adult owl and squirrel monkeys. Neuroscience 10:639–641.

Merzenich MM, Kaas JH, Wall J, Nelson RJ, Sur M, Felleman D. 1983b. Topographic reorganization of somatosensory cortical areas 3b and 1 in adult monkeys following restricted deafferentation. Neuroscience 8:33–55.

Merzenich M, Nelson R, Stryker M, Cynader M, Schoppmann A, Zook J. 1984. Somatosensory cortical map changes following digit amputation in adult monkeys. J Comp Neurol 224:591–605.

Murray GM, Sessle BJ. 1992. Functional properties of single neurons in the face primary motor cortex of the primate. II. Relations with trained orofacial motor behavior. J Neurophysiol 67:759–774.

Murray GM, Lin LD, Moustafa EM, Sessle BJ. 1991. Effects of reversible inactivation by cooling of the primate face motor cortex on the performance of a trained tongue-protrusion task and a trained biting task. J Neurophysiol 65:511–530.

Nelson RJ, Kaas JH. 1981. Connections of the ventroposterior nucleus of the thalamus with the body surface representations in cortical areas 3b and 1 of the cynomolgus macaque (*Macaca fascicularis*). J Comp Neurol 199:29–64.

Nelson RJ, Sur M, Felleman DJ, Kaas JH. 1980. Representations of the body surface in postcentral parietal cortex of *Macaca fascicularis*. J Comp Neurol 192:611–643.

Ogawa H. 1994. Gustatory cortex of primates: anatomy and physiology. Neurosci Res 20:1–13.

Ogawa H, Ito S-I, Nomura T. 1985. Two distinct projection areas from tongue nerves in the frontal operculum of macaque monkeys as revealed with evoked potential mapping. Neurosci Res 2:447–459.

Ogawa H, Ito S-I, Nomura T. 1989. Oral cavity representation at the frontal operculum of macaque monkeys. Neurosci Res 6:283–298.

Olucha F, Martonez-Garcia F, Lopez-Garcia C. 1985. A new stabilizing agent for the tetramethyl benzidine (TMB) reaction product in the histochemical detection of horseradish peroxidase (HRP). J Neurosci Methods 13:131–138.

Padberg J, Cerkevich C, Engle J, Rajan AT, Recanzone G, Kaas J, Krubitzer L. 2009. Thalamocortical connections of parietal somatosensory cortical fields in macaque monkeys are highly divergent and convergent. Cereb Cortex 19: 2038–2064.

Pearson RCA, Powell TPS. 1978. The cortico-cortical connections to area 5 of the parietal lobe from the primary somatic snsory cortex of the monkey. Proc R Soc Lond B Biol Sci 200:103–108.

Pearson RCA, Powell TPS. 1985. The projection of the primary somatic sensory cortex upon area 5 in the monkey. Brain Res Rev 9:89–107.

Petrides M, Tomaiuolo F, Yeterian EH, Pandya DN. 2012. The prefrontal cortex: comparative architectonic organization in the human and the macaque monkey brains. Cortex 48:46–57.

Plata-Salamán CR, Scott TR. 1992. Taste neurons in the cortex of the alert cynomolgus monkey. Brain Res Bull 28: 333–336.

Pons TP, Kaas JH. 1985. Connections of area 2 of somatosensory cortex with the anterior pulvinar and subdivisions of the ventroposterior complex in macaque monkeys. J Comp Neurol 240:16–36.

Pons TP, Kaas JH. 1986. Corticocortical connections of area 2 of somatosensory cortex in macaque monkeys: a correlative anatomical and electrophysiological study. J Comp Neurol 248:313–335.

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Pons TP, Garraghty PE, Cusick CG, Kaas JH. 1985a. A sequential representation of the occiput, arm, forearm and hand across the rostrocaudal dimension of areas 1, 2 and 5 in macaque monkeys. Brain Res 335:350–353.

Pons TP, Garraghty PE, Cusick CG, Kaas JH. 1985b. The somatotopic organization of area 2 in macaque monkeys. J Comp Neurol 241:445–466.

Pons TP, Garraghty PE, Ommaya AK, Kaas JH, Taub E, Mishkin M. 1991. Massive cortical reorganization after sensory deafferentation in adult macaques. Science 252:1857–1860.

Pritchard T, Norgren R. 2004. Gustatory sysytem. In: The human nervous system, 2nd ed. Amsterdam: Elsevier. p 1171–1198.

Pritchard T, Hamilton R, Morse J, Norgren R. 1986. Projections of thalamic gustatory and lingual areas in the monkey, *Macaca fascicularis*. J Comp Neurol 244:213–228.

Qi H-X, Kaas J. 2004. Myelin stains reveal an anatomical framework for the representation of the digits in somatosensory area 3b of macaque monkeys. J Comp Neurol 477:172–187.

Qi H-X, Stepniewska I, Kaas JH. 2000. Reorganization of primary motor cortex in adult macaque monkeys with long-standing amputations. J Neurophysiol 84:2133–2147.

Qi H-X, Lyon DC, Kaas JH. 2002. Cortical and thalamic connections of the parietal ventral somatosensory area in marmoset monkeys (*Callithrix jacchus*). J Comp Neurol 443:168–182.

Rolls E. 2000. The orbitofrontal cortex and reward. Cereb Cortex 10:284–294.

Rolls E, Baylis L. 1994. Gustatory, olfactory, and visual convergence within the primate orbitofrontal cortex. J Neurosci 14:5437–5452.

Rolls ET, Sienkiewicz ZJ, Yaxley S. 1989. Hunger modulates the responses to gustatory stimuli of single neurons in the caudolateral orbitofrontal cortex of the macaque monkey. Eur J Neurosci 1:53–60.

Sanides F. 1968. The architecture of the cortical taste nerve areas in squirrel monkey (*Saimiri sciureus*) and their relationships to insular, sensorimotor and prefrontal regions. Brain Res 8:97–124.

Schmued L, Slikker W. 1999. Black-Gold: a simple, high-resolution histochemical label for normal and pathological myelin in brain tissue sections. Brain Res 837:289–297.

Scott TR, Plata-Salamán CR. 1999. Taste in the monkey cortex. Physiol Behav 67:489–511.

Scott TR, Yaxley S, Sienkiewicz ZJ, Rolls ET. 1986. Gustatory responses in the frontal opercular cortex of the alert cynomolgus monkey. J Neurophysiol 56:876–890.

Seelke AMH, Padberg JJ, Disbrow E, Purnell SM, Recanzone G, Krubitzer L. 2012. Topographic maps within Brodmann's area 5 of macaque monkeys. Cereb Cortex 22:1834–1850.

Shepherd GM. 2012. Neurogastronomy: how the brain creates flavor and why it matters. New York: Columbia University Press.

Smith-Swintosky VL, Plata-Salaman CR, Scott TR. 1991. Gustatory neural coding in the monkey cortex: stimulus quality. J Neurophysiol 66:1156–1165.

Stepniewska I, Preuss TM, Kaas JH. 2006. Ipsilateral cortical connections of dorsal and ventral premotor areas in New World owl monkeys. J Comp Neurol 495:691–708.

Tataranni PA, Gautier J-F, Chen K, Uecker A, Bandy D, Salbe AD, Pratley RE, Lawson M, Reiman EM, Ravussin E. 1999. Neuroanatomical correlates of hunger and satiation in humans using positron emission tomography. Proc Natl Acad Sci U S A 96:4569–4574.

Toda T, Taoka M. 2001. The complexity of receptive fields of periodontal mechanoreceptive neurons in the postcentral area 2 of conscious macaque monkey brains. Arch Oral Biol 46:1079–1084.

Toda T, Taoka M. 2002a. Hierarchical somesthetic processing of tongue inputs in the postcentral somatosensory cortex of conscious macaque monkeys. Exp Brain Res 147:243–251.

Toda T, Taoka M. 2002b. Integration of the upper and lower lips in the postcentral area 2 of conscious macaque monkeys (*Macaca fuscata*). Arch Oral Biol 47:449–456.

Toda T, Taoka M. 2004. Converging patterns of inputs from oral structures in the postcentral somatosensory cortex of conscious macaque monkeys. Exp Brain Res 158:43–49.

Veenman CL, Reiner A, Honig MG. 1992. Biotinylated dextran amine as an anterograde tracer for single- and double-labeling studies. J Neurosci Methods 41:239–254.

Watanabe M. 1996. Reward expectancy in primate prefrental neurons. Nature 382:629–632.

Wu CWH, Kaas JH. 2003. Somatosensory cortex of prosimian galagos: physiological recording, cytoarchitecture, and corticocortical connections of anterior parietal cortex and cortex of the lateral sulcus. J Comp Neurol 457:263–292.

Yeterian EH, Pandya DN, Tomaiuolo F, Petrides M. 2012. The cortical connectivity of the prefrontal cortex in the monkey brain. Cortex 48:58–81.

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